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Supplementary information

Metal–organic frameworks embedded in a liposome facilitate overall photocatalytic water splitting

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Metal–organic frameworks embedded in a liposome facilitate overall photocatalytic water splitting

Huihui Hu,[†] Zhiye Wang,[†] Lingyun Cao,[†] Lingzhen Zeng,[†] Cankun Zhang,[†] Cheng Wang[*,†] and Wenbin Lin[‡]

[†]Collaborative Innovation Center of Chemistry for Energy Materials, State Key Laboratory of Physical Chemistry of Solid Surfaces, Department of Chemistry, College of Chemistry and Chemical Engineering, Xiamen University, Xiamen 361005, P.R. China

[‡]Department of Chemistry, The University of Chicago, 929 East 57th Street, Chicago, Illinois 60637, United States

Email: wangchengxmu@xmu.edu.cn

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1. General Experimental and Chemicals

Chemicals. Biphenyl-4,4'-dicarboxylic acid (H_2bpdc, 97%), zirconium(IV) chloride (99.9%, $ZrCl_4$), hafnium(IV) chloride (99.9%, $HfCl_4$), iridium(III) chloride ($IrCl_3 \cdot 3H_2O$), and formic acid (HCO_2H, 99%) were purchased from J&K Scientific Co., Ltd. N,N'-dimethylformamide (DMF) was purchased from Sinopharm Chemical Reagent Co. Ar (99.999%) was purchased from Fuzhou Xinhang Industrial gases Co., Ltd. Meso-Tetra(4-carboxyphenyl)porphine (TCPP, 97%) was purchased from Zhengzhou Ruke Biological Co., Ltd.

General experimental. Reagents were commercially available and used without further purification unless otherwise indicated. Thermogravimetric analysis (TGA) was performed in air using a Shimadzu TGA-50 equipped with an alumina pan and heated at a rate of 5 °C per minute. The powder X-ray diffraction data were collected on Agilent SuperNova, Rigaku and Bruker D8 Venture diffractometers using Cu Kα radiation sources (λ = 1.54178 Å). Transmission electron microscopy images were obtained on JEOL 2100 and Tecnai F20 High Resolution Transmission Electron Microscopy and JEOL 1400. Before performed the TEM characterization, we fishing the samples with copper grid. AFM images were taken on a Bruker Multimode V, using Tapping Mode, scan speed is 1Hz, and before performed the characterization, the samples were dripped directly onto clean mica flakes. ^{1}H NMR spectra were recorded on a Bruker AV-500 and referenced to the proton resonance resulting from incomplete deuteration of the DMSO-d_6 (δ 2.50). DLS measurements were performed with a Malvern Zetasizer Nano ZS (Malvern, Herrenberg, Germany). Fluorescence spectra were taken at room temperature on Hitachi FL980 and Hitachi F7000. The emission wavelength selection is achieved through the grating-based monochromator. The confocal fluorescence microscopy images were performed with a Leica TCS SP5 and Leica TCS SP8. Using 100x Oil lens to observe the sample and for LP-HER-MOF, the excited laser is 405 nm, for LP-WOR-MOF , the excited laser is 458 nm, for LP-HER-WOR-MOF, in HER-MOF channel, the excited laser is 552nm, in WOR-MOF channel, the excited laser is 488nm, before performed the characterization, we directly transfer the sample to the confocal dish, and slide diameter is 15 mm. The femtosecond transient absorption system is based on Coherent's regeneratively amplified titanium-doped gemstone femtosecond laser (center wavelength 800 nm, pulse width 35 fs, exit light intensity 5.8 mJ/pulse, repetition rate 1 kHz), optical parametric amplifier and home-built delay and detection systems with Andor spectrometers. Gas phasic products were quantified by gas chromatography with a thermal conductivity detector (TCD, column: TDX-01 and 5Å molecular sieve, GC7900).

2. Ligand synthesis

Ligand synthesis. [(H_4TCPP)Zn] was synthesized as reported previously[1] and [(H_4TCPP)Pt] was synthesized using a similar procedure.

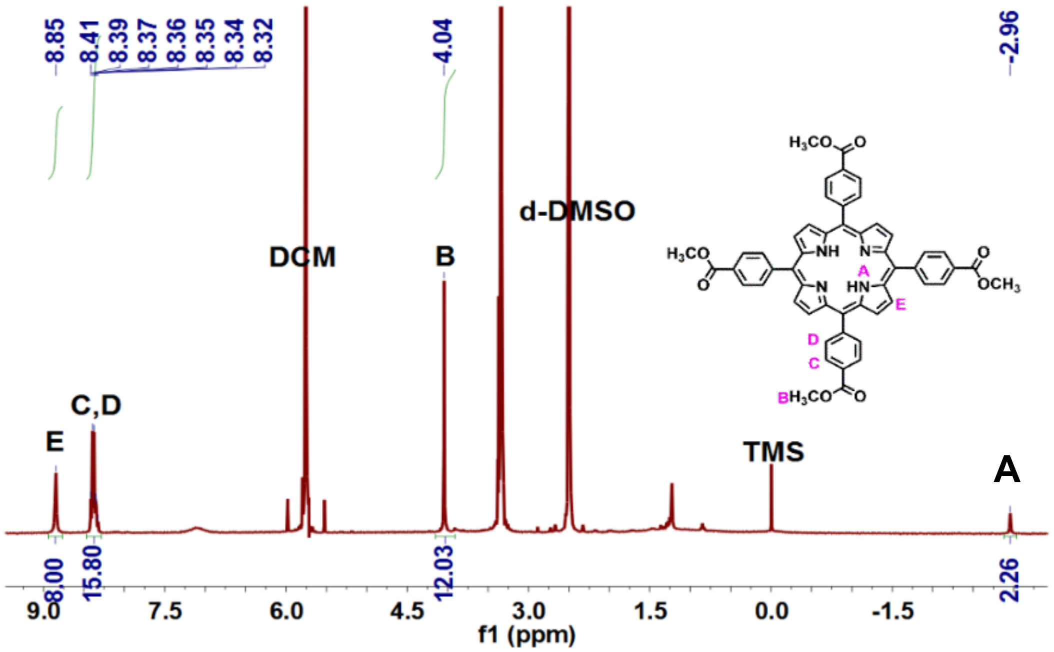
Synthesis of 5,10,15,20-tetrakis(4-methoxycarbonylphenyl)porphyrin (Me_4TCPP) In a 250 mL two-neck round bottom flask in an ice water bath, meso-tetra(4-carboxyphenyl)porphine (1.0 g, 1.26 mmol) was dissolved in methanol (150 mL) before 3mL of concentrated sulfuric acid was added. The solution was refluxed for 12h in dark. After the reaction mixture was cooled to room temperature, the pH was adjusted to 7 with an ammonia solution, and excess methanol was removed by rotary evaporation. The aqueous phase was then extracted with dichloromethane and water (v:v = 1:1), which was dried by rotary evaporation to obtain the pure product. (952 mg, 1.134 mmol, 90% yield). ^{1}H NMR (400 MHz, DMSO-d_6) δ 8.85 (s, 8H), 8.40 (d, 8H), 8.35(d, 8H), 4.04 (s, 12H), -2.96 (s, 2H). (Supplementary Figure 1)

Synthesis of 5,10,15,20-tetrakis(4-carboxyphenyl)porphyrinato-Zn ([(H_4TCPP)Zn]). Me_4TCPP (0.854 g, 1.0 mmol) and $Zn(NO_3)_2{\cdot}6H_2O$ (3.8 g, 12.8 mmol) were added to 100 mL of DMF and the mixture was refluxed for 12 h. After cooling to room temperature, 150 mL of H_2O was introduced. The resultant precipitate was filtered and washed with 50 mL of H_2O for three times. The resultant solid was dissolved in CH_2Cl_2, washed three times with 1 M HCl and twice with water. The organic layer was then dried over anhydrous sodium sulfate and evaporated to give a dark brown solid (720 mg, 0.8 mmol, 80% yield). The as-synthesized ester (0.72 g, 0.8 mmol) was stirred in a mixed solvent of THF (25 mL) and MeOH (25 mL), to which a solution of KOH (2.63 g, 46.95 mmol) in H_2O (25 mL) was introduced. This mixture was refluxed for 12 h. After cooling to room temperature, THF and MeOH were evaporated. Additional water was added to the resultant aqueous phase. Then the solution was acidified with 1 M HCl until no further precipitate was detected. The brown solid was collected by filtration, washed with water and dried in vacuum. (610 mg, 0.72 mmol, 90% yield). ^{1}H NMR (400 MHz, DMSO-d_6) δ 8.80 (s, 8H), 8.35 (d, 8H), 8.28 (d, 8H). (Supplementary Figure 2)

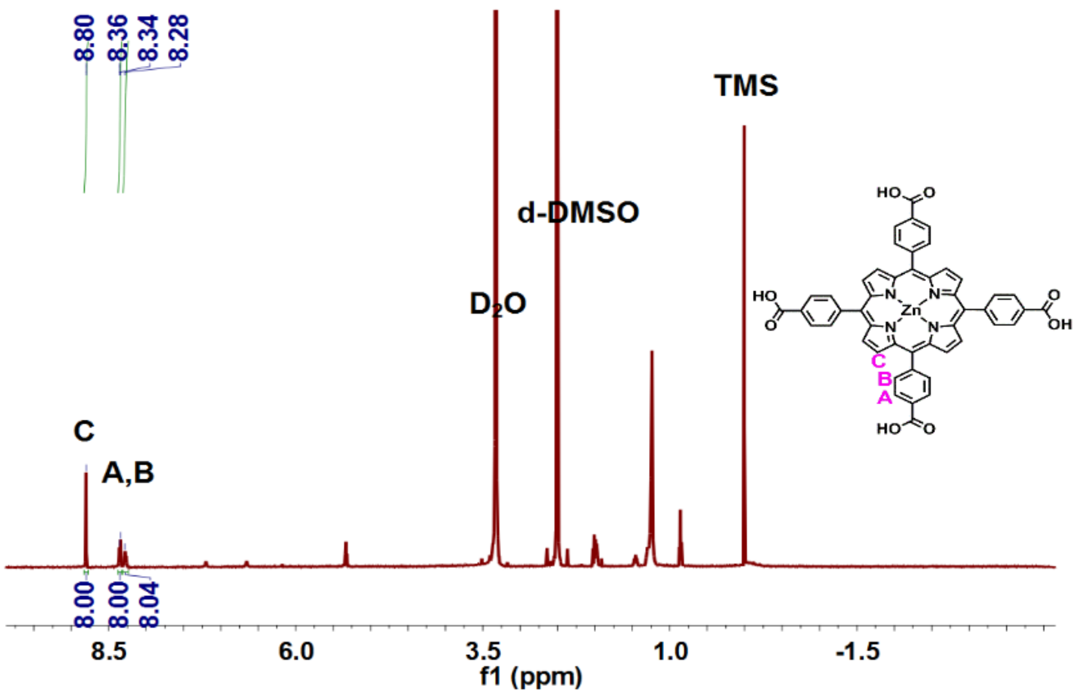
Synthesis of 5,10,15,20-tetrakis(4-carboxyphenyl)porphyrin-Pt ([(H_4TCPP)Pt]). Me_4TCPP (100 mg, 0.12 mmol) and K_2PtCl_4 (70 mg, 0.21 mmol) were added into 30 mL of benzonitrile and the mixture was refluxed for 24 h under Ar atmosphere in dark. After cooling to room temperature, the solvent was removed by rotary evaporation on an oil bath. The solid was then extracted with dichloromethane and water (v:v = 1:1), and the organic phase was dried by rotary evaporation to afford the product (92.8 mg, 0.09 mmol, 75% yield). The as-synthesized ester (92.8 mg, 0.09 mmol) was stirred in a mixed solvent of THF (25 mL) and MeOH (25 mL), to which a solution of KOH (2.63 g, 46.95 mmol) in H_2O (25 mL) was introduced. This mixture was refluxed for 12 h. After cooling to room temperature, THF and MeOH were evaporated. Additional water was added to the resulting aqueous phase. Then the solution was acidified with 1 M HCl until no further precipitate was detected. The red solid was collected by filtration, washed with water and dried in vacuum. (84 mg, 0.086 mmol, 95% yield). ^{1}H NMR (400 MHz, DMSO-d_6) δ 8.76 (s, 8H), 8.37 (d, 8H), 8.29(d, 8H). (Supplementary Figure 3)

Synthesis of [Ru^{II}($BPY)_2$(H_2BPYDC)]Cl_2 ([Ru]$^{2+}$). [Ru]$^{2+}$ was synthesized as reported previously.[2-3] The Ru(II) precursor cis-[$(BPY)_2RuCl_2$] was prepared by reacting 2 equiv. of

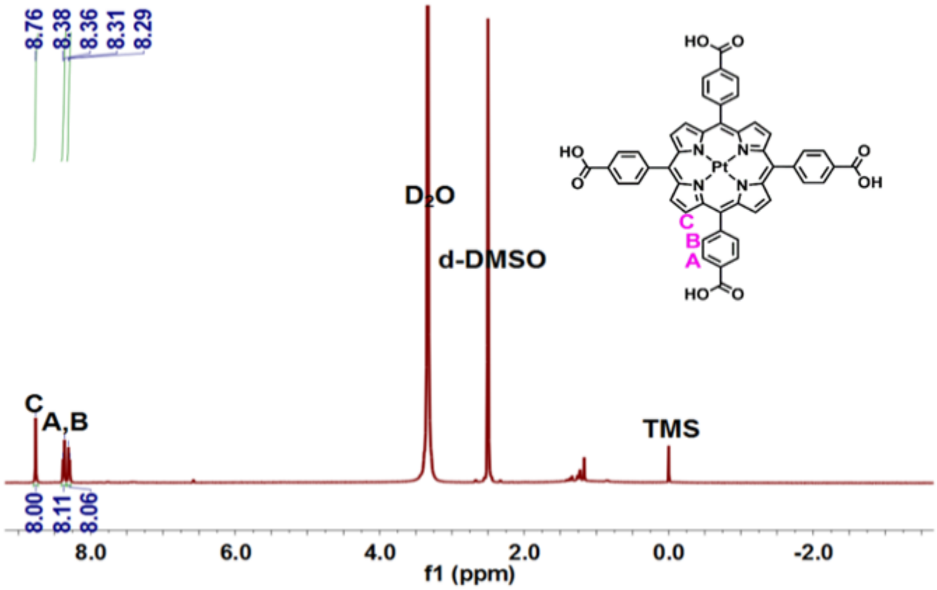
2,2'-bipyridine with $RuCl_3 \cdot 3H_2O$ in dimethylformamide (DMF) under reflux for 24 h to obtain the reddish brown product. Then the *cis*-$[Ru(BPY)_2Cl_2]$ (160 mg, 0.33 mmol) was mixed with 2,2'-bipyridine-5,5'-dicarboxylic acid (101 mg, 0.42 mmol) in 20 mL of $EtOH/H_2O$, refluxed for 12 hours under Argon before being concentrated. The solid was recrystallized from MeOH-diethyl ether. ^1H NMR (400 MHz, DMSO-d_6) δ 8.76 (s, 8H), 8.37 (d, 8H), 8.29(d, 8H). (Supplementary Figure 4)



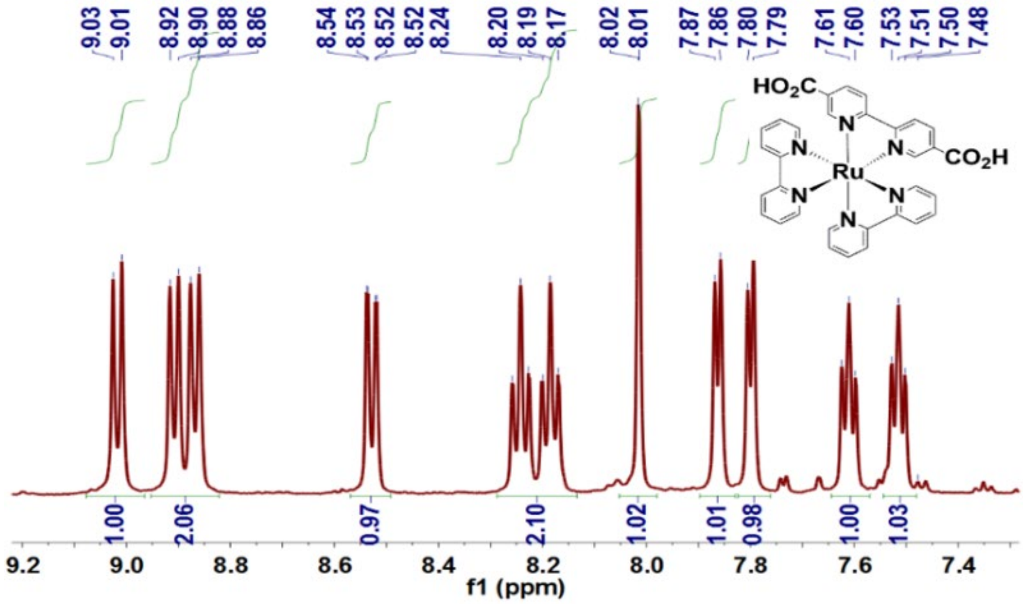
Supplementary Figure 1. ^1H-NMR spectrum of Me_4TCPP in d_6-DMSO.



Supplementary Figure 2. ^1H-NMR spectrum of $[(H_4TCPP)Zn]$ in d_6-DMSO.

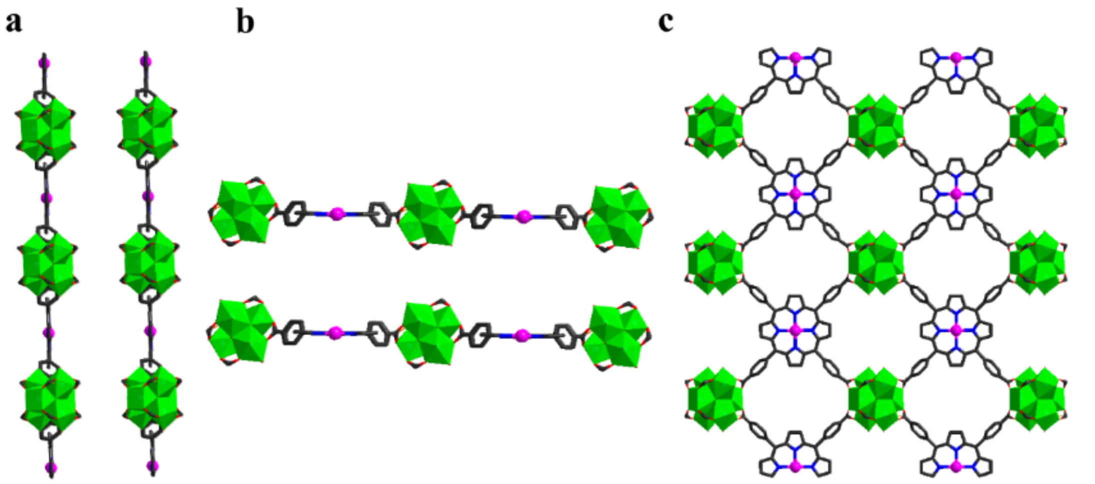


Supplementary Figure 3. ^{1}H-NMR spectrum of [$(H_4TCPP)Pt$] in d_6-DMSO.

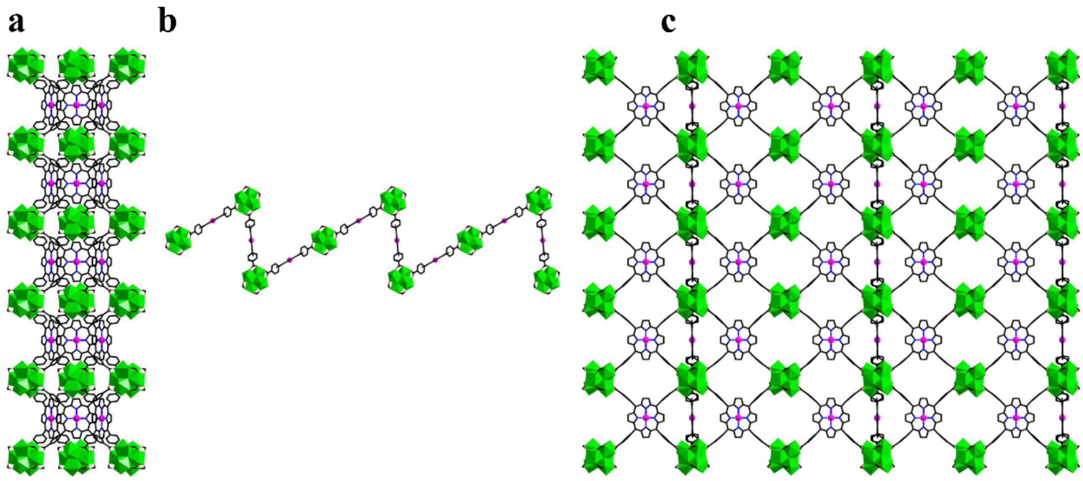


Supplementary Figure 4. ^{1}H-NMR spectrum of $[Ru]^{2+}$ in d_6-DMSO.

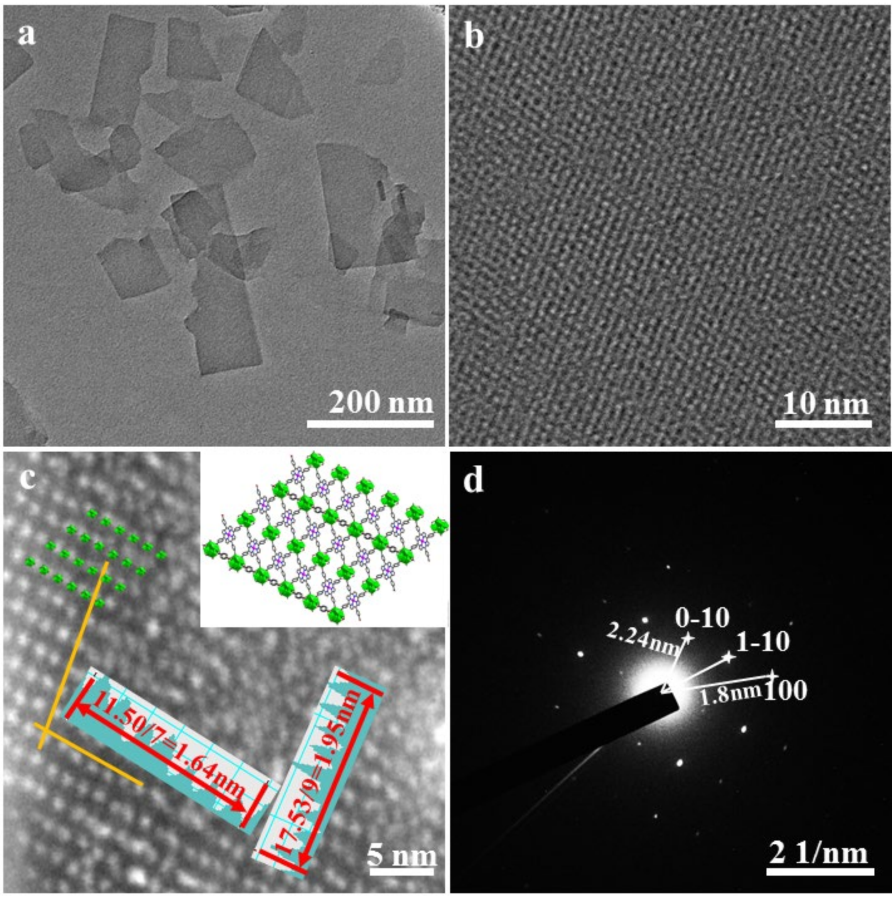
3. Synthesis and Characterization of HER-MOF



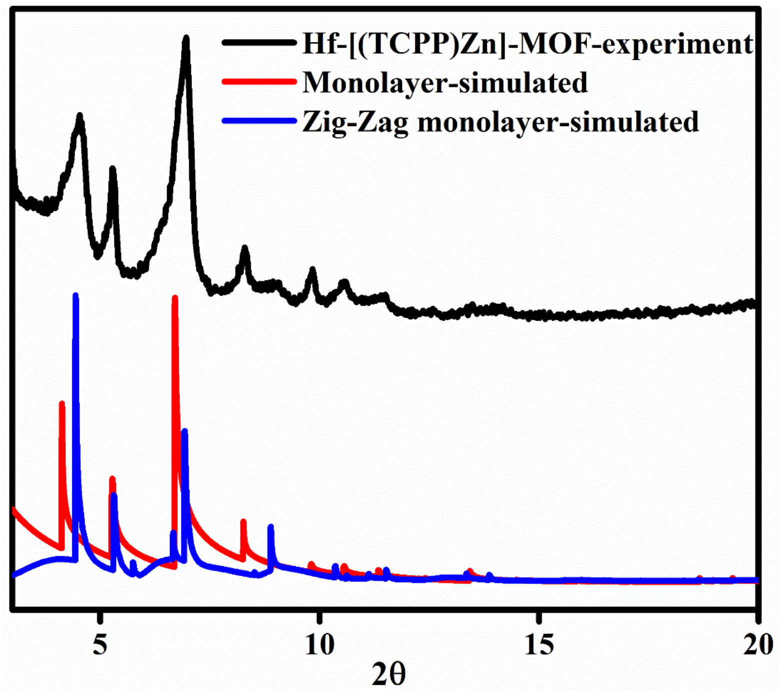
Supplementary Figure 5. The structure of the $[Hf_6(\mu_3\text{-}O)_4(\mu_3\text{-}OH)_4(\mu_1\text{-}OH)_2(\mu_1\text{-}H_2O)_2(HCO_2)_6([(TCPP)Zn])]$: viewed along the *a* axis(a). viewed along the *b* axis (b). topview of the monolayer (c).



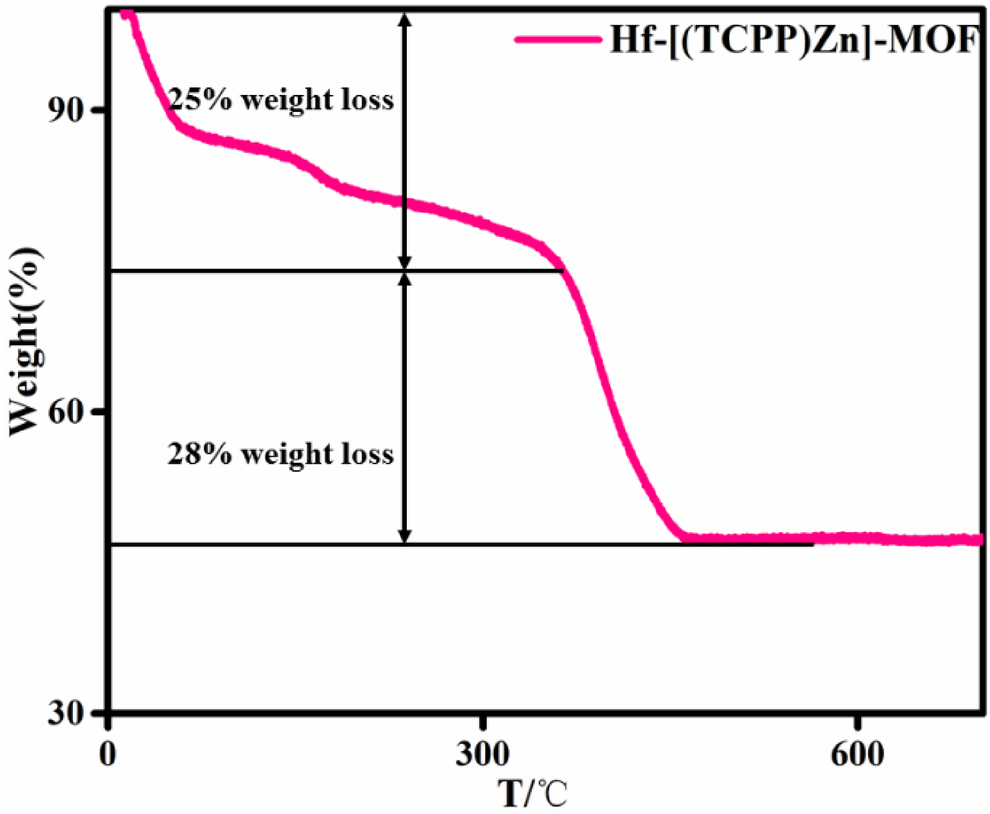
Supplementary Figure 6. The structure of the zig-zag monolayer: viewed along the *a* axis (a). viewed along the *b* axis (b). topview of the monolayer(c).



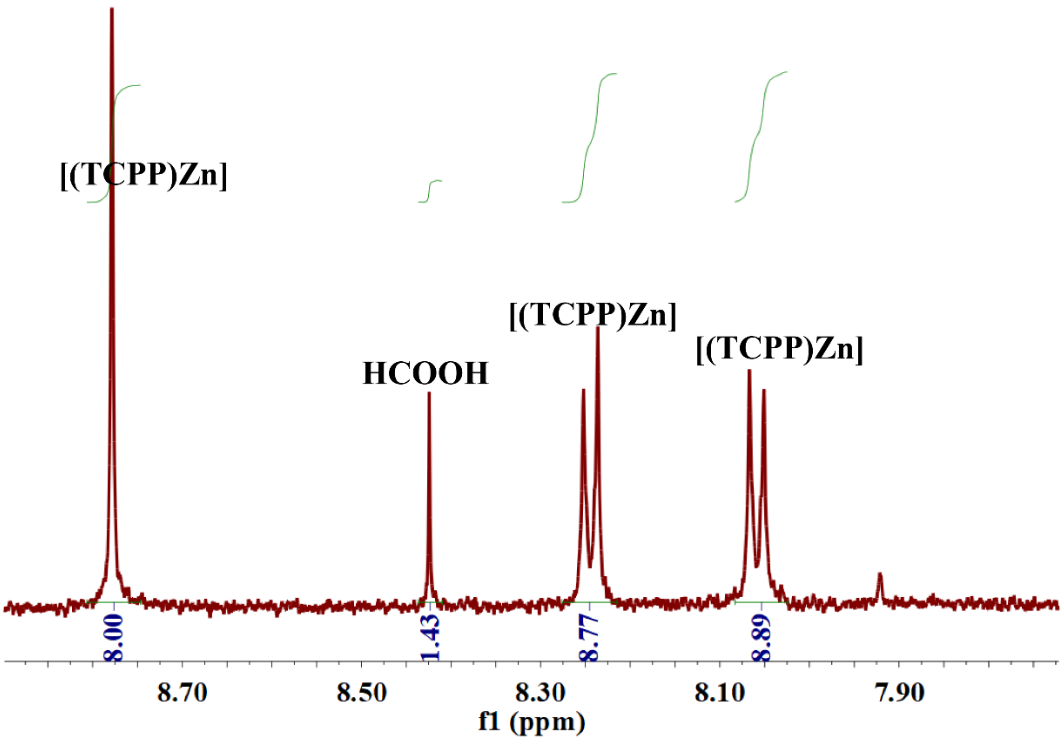
Supplementary Figure 7. TEM image of the Hf-[(TCPP)Zn]-MOF with a structural model superimposed on the image(a) & (b); The HAADF image of Hf-[(TCPP)Zn]-MOF(c); SAED image of Hf-[(TCPP)Zn]-MOF(d).



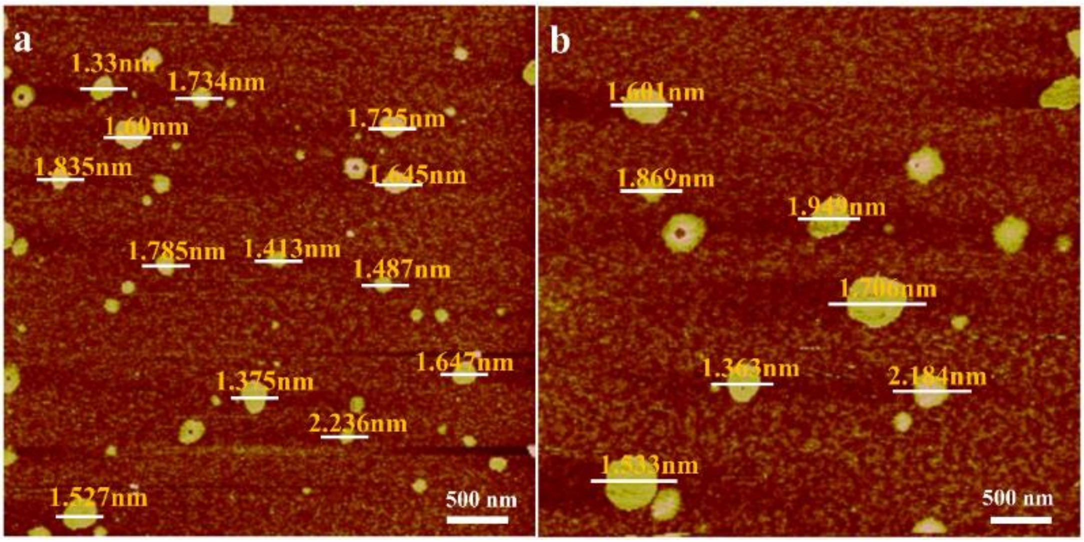
Supplementary Figure 8. The PXRD of Hf-[(TCPP)Zn]-MOF, together with the simulated patterns from the structural models of Hf-[(TCPP)Zn]-MOF of a flat monolayer and a zig-zag monolayer.



Supplementary Figure 9. The TGA of Hf-[(TCPP)Zn]-MOF.

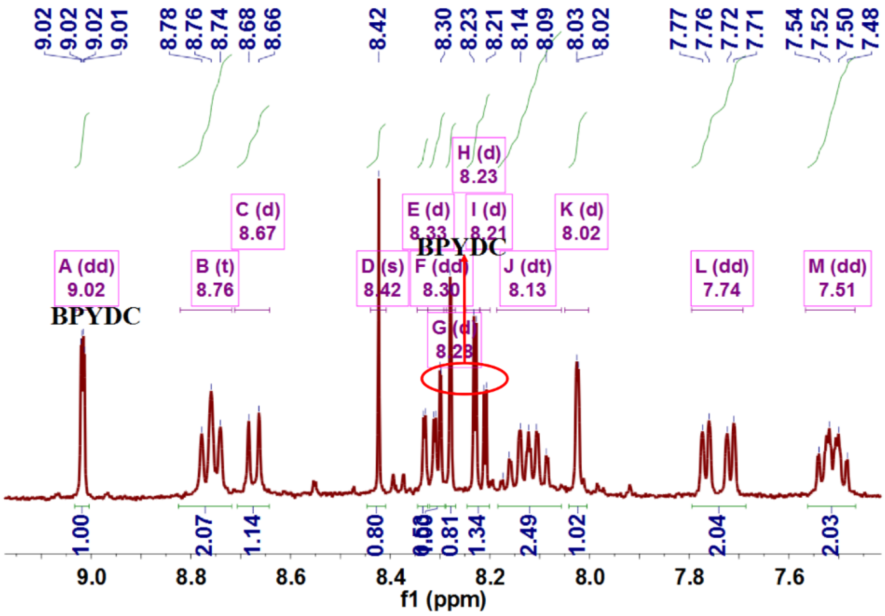


Supplementary Figure 10. The ^{1}H NMR spectrum of the dissolved Hf-[(TCPP)Zn]-MOF. The procedure for taking the spectrum is as following: As-synthesized Hf-[(TCPP)Zn]-MOFs were washed with DMF and acetone several times and dried in vacuum at 50 °C; The dried MOFs were dissolved with a solution of 0.1 mL of D_3PO_4 and 0.4 mL of d_6-DMSO.

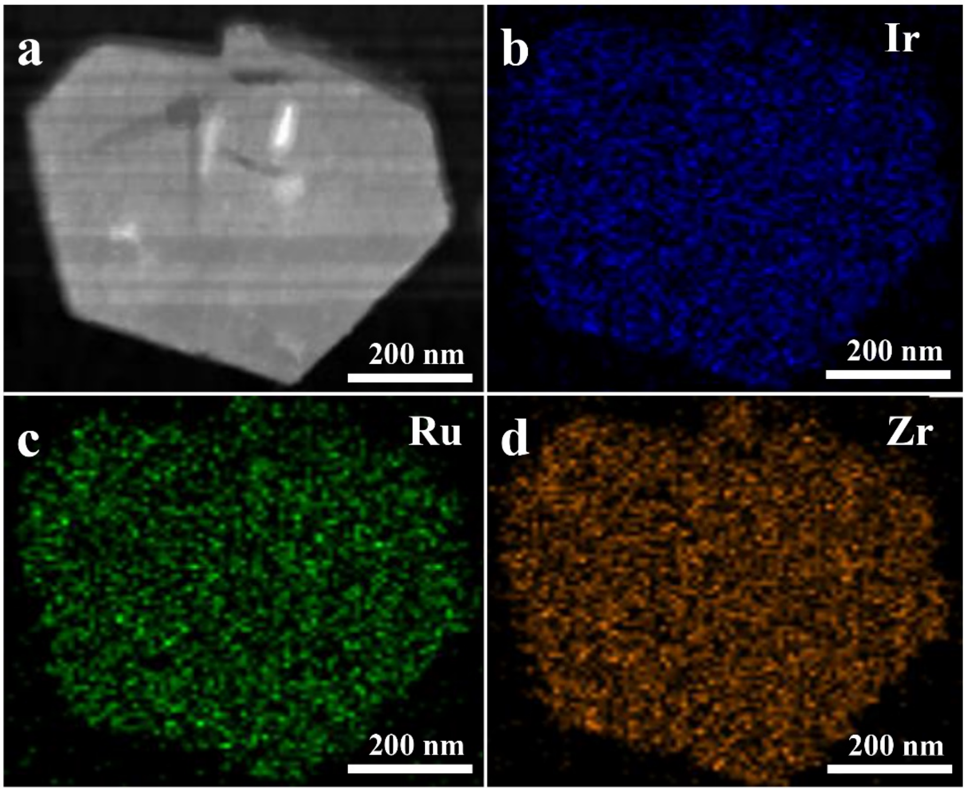


Supplementary Figure 11. The AFM images of the HER-MOF

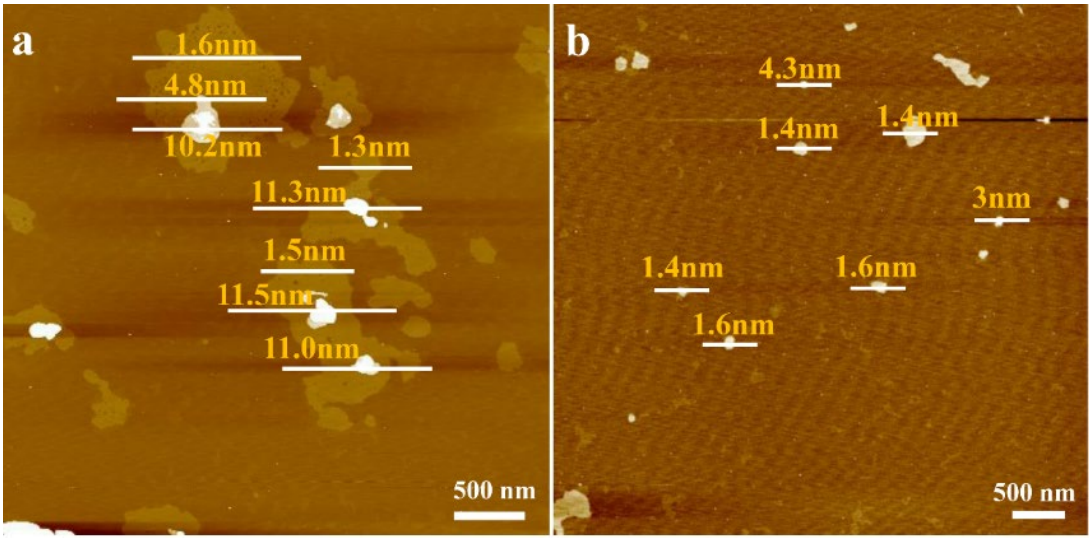
4. Synthesis and Characterization of WOR-MOF



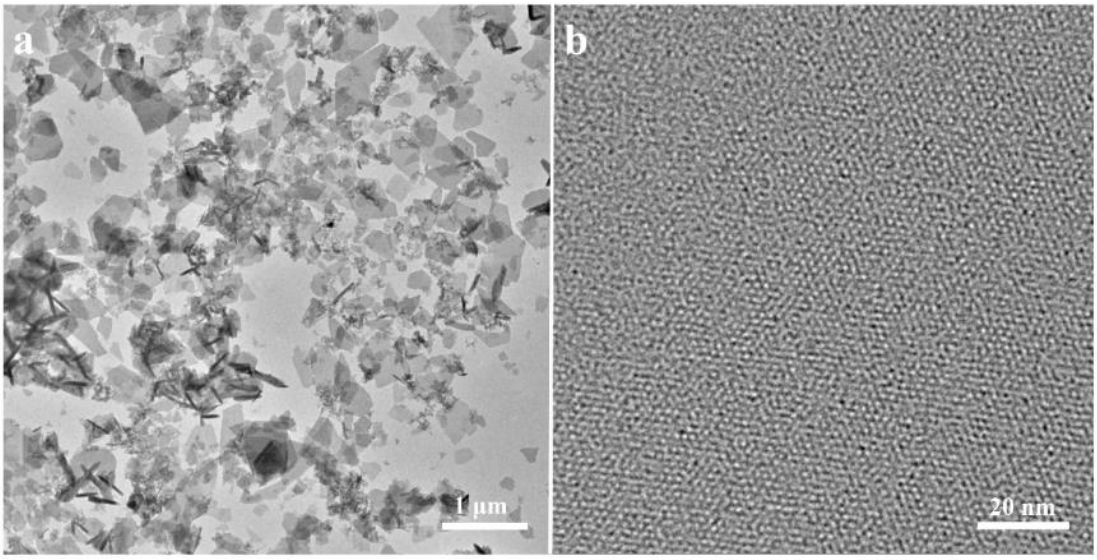
Supplementary Figure 12. ^{1}H NMR spectrum of the dissolved WOR-MOF. The procedure for taking the spectrum is as following: As-synthesized WOR-MOFs were washed with DMF and acetone several times and dried in vacuum at 50 °C; The dried MOFs were dissolved with a solution of 0.1 mL of D_3PO_4 and 0.4 mL of d_6-DMSO.



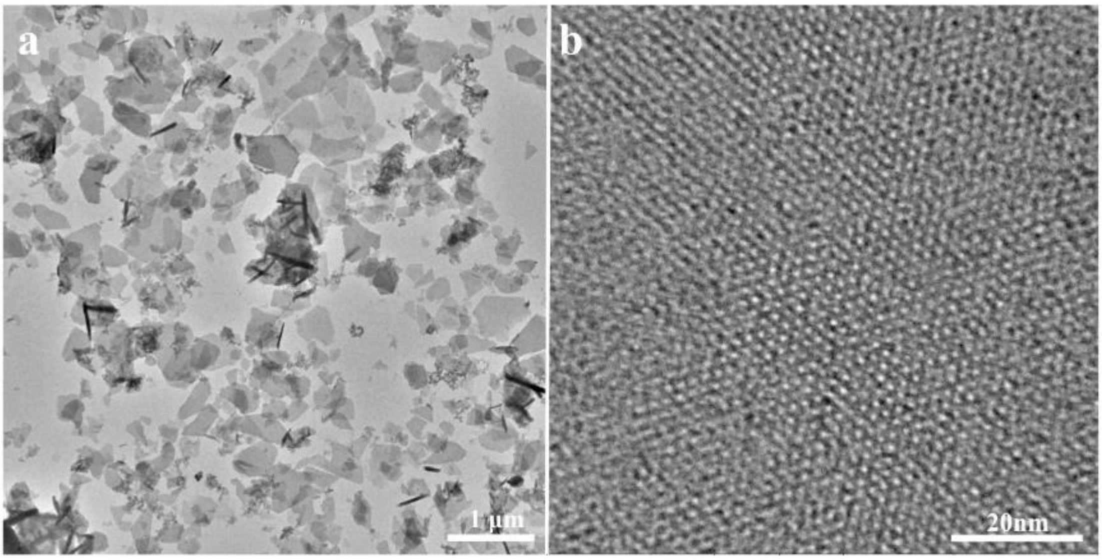
Supplementary Figure 13. TEM EDX Mapping of the WOR-MOF.



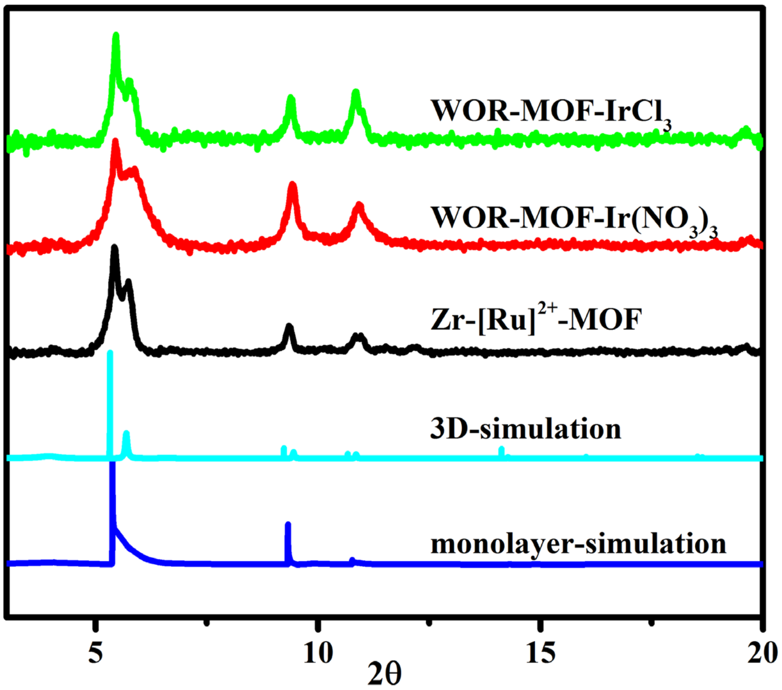
Supplementary Figure 14. AFM images of WOR-MOF (a&b).



Supplementary Figure 15. TEM (a) and HRTEM (b) images of Zr-$[Ru]^{2+}$-MOF.



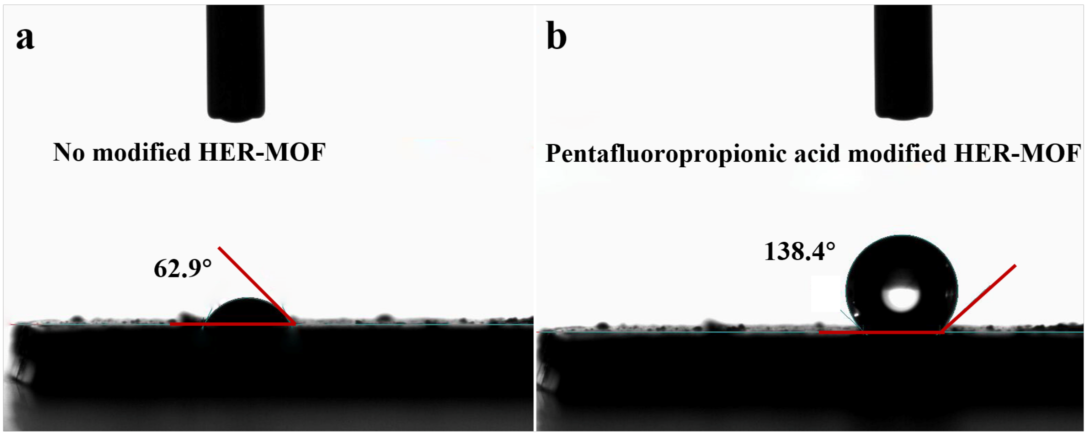
Supplementary Figure 16. TEM (a) and HRTEM (b) images of WOR-MOF.



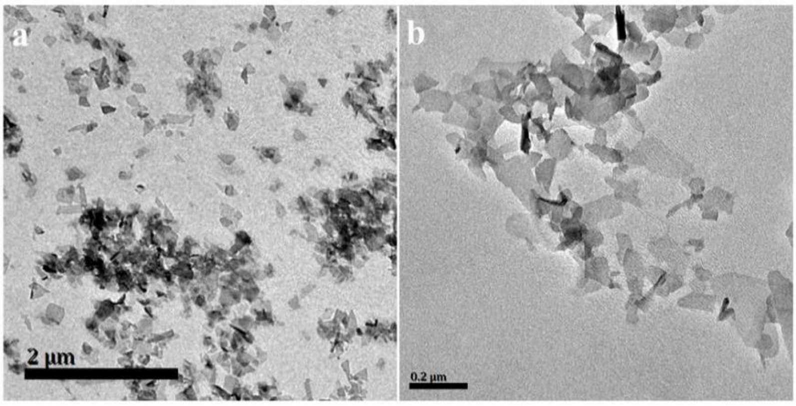
Supplementary Figure 17. PXRD pattern of WOR-MOF compared with the simulated ones.

5. Synthesis and Characterization of LP-MOF Hybrids

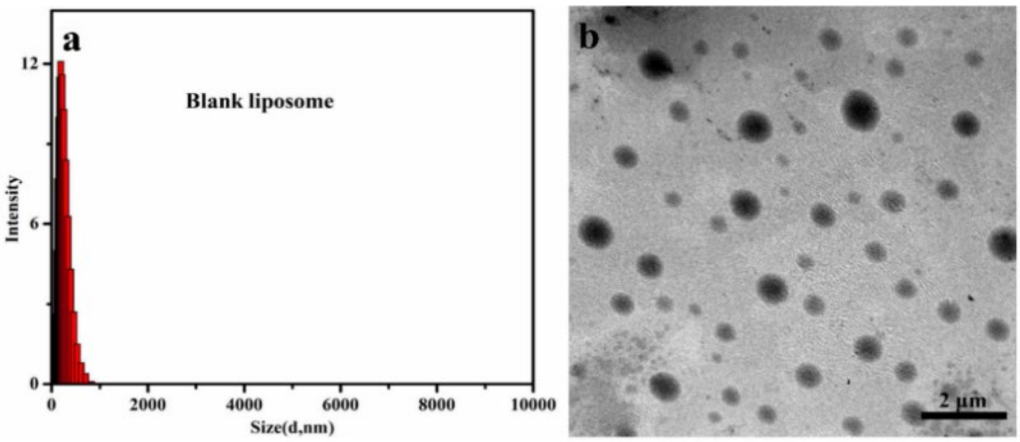
Synthesis and characterization of blank liposome. The liposomes were synthesized following similar procedures in the literature by film dispersion method.[4] The Lecithin (PC, 1709 μL) and cholesterol (Chol, 291 μL) dissolved in chloroform were dried into a film on the wall of a flask and kept in vacuum overnight. The dried lipid film was then hydrated in the presence of PYREXTM glass beads in dark. The mixture was then ultra-sonicated for 10 min, followed by 10 freeze-thaw cycles using liquid N_2, and then immersed in a 55 °C water bath to form a multilamellar vesicle suspension. The suspension was slowly cooled to room temperature in 30 min. The suspension was then extruded with a mini-extruder (Avanti Polar Lipids) at room temperature using a polycarbonate membrane with a pore size of 1 μm for 11 cycles.



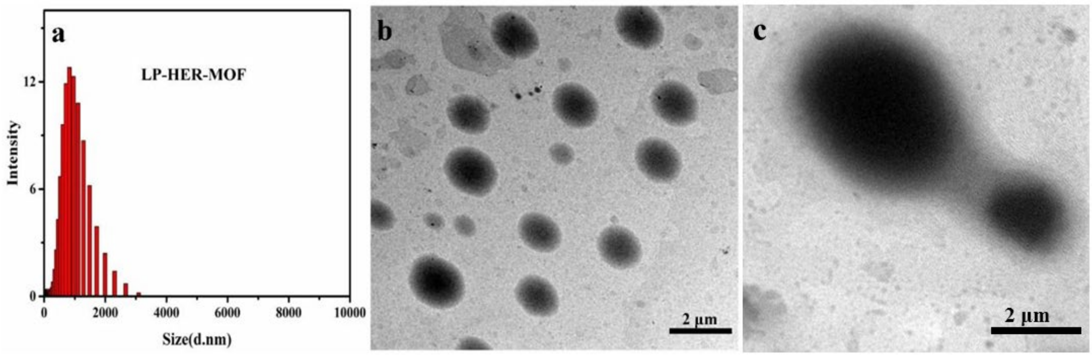
Supplementary Figure 18. The micrographs of water droplets on HER-MOF(a) and Micrographs of water droplets of pentafluoropropionic acid modified HER-MOF(b).



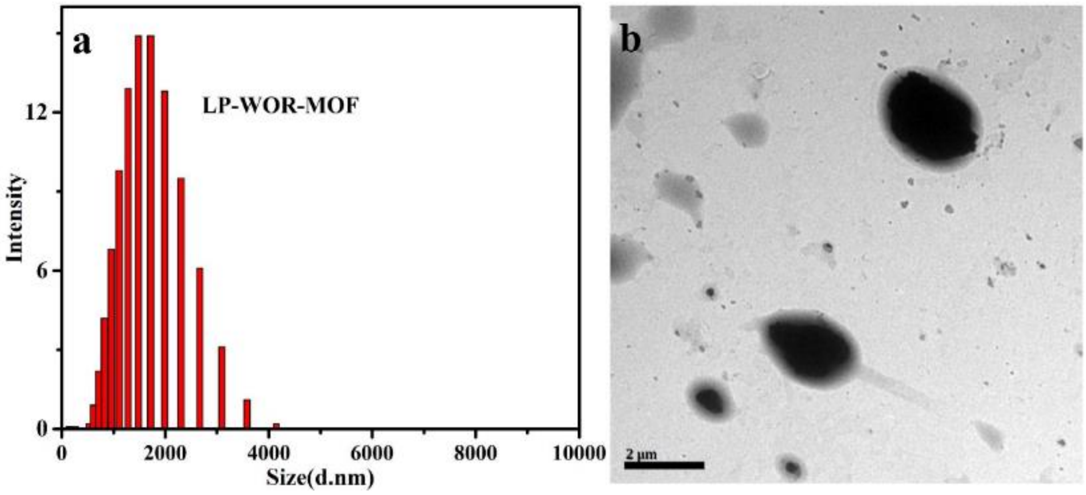
Supplementary Figure 19. The TEM images of HER-MOFs that were extracted from the LP-HER-MOF assemblies (a&b). We added DMF to the liposome suspension to destroy the liposome and dissolve the lipids. The MOFs were then obtained by centrifugation (Speed 9600 r/min, ~6000 g, time: 5min), and the amounts of MOFs are determined by ICP-MS.



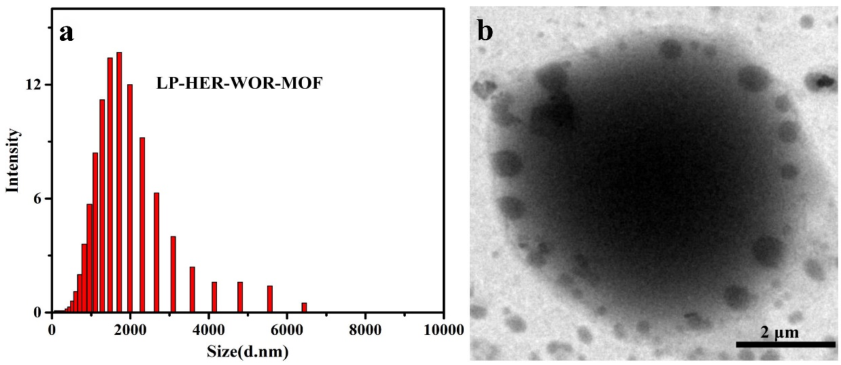
Supplementary Figure 20. Dynamic light scattering measurement of blank liposome (a) and TEM of blank liposome (b).



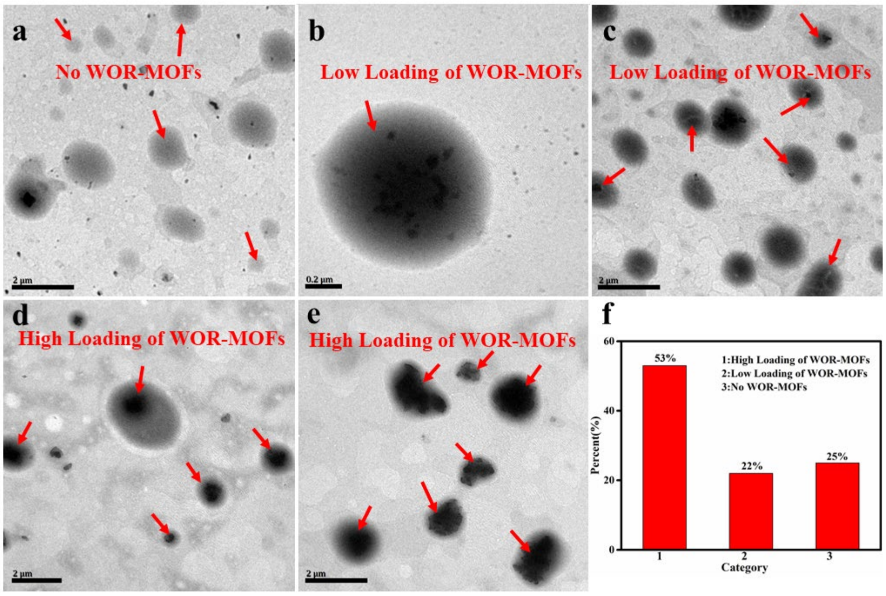
Supplementary Figure 21. Dynamic light scattering measurement of LP-HER-MOF (a) and TEM of LP-HER-MOF (b)&(c). The size of the liposome with MOF are typically significantly larger than 1 μm. We attributed this to the fusion of different liposomes. We frequently observed connected liposomes that are about to fuse under TEM (c). The addition of the MOF in the lipid bilayer may disturb its structure to enhance its fluidity that facilitates fusion.



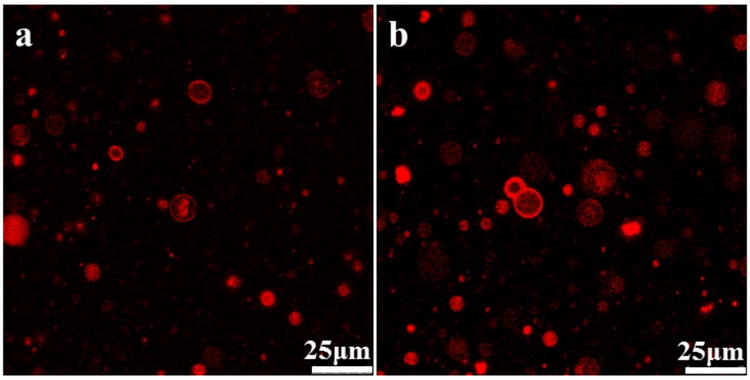
Supplementary Figure 22. Dynamic light scattering measurement of LP-WOR-MOF (a) and TEM of LP-WOR-MOF (b).



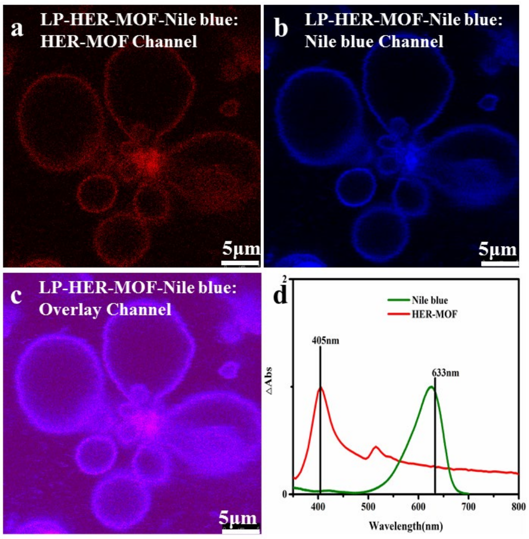
Supplementary Figure 23. Dynamic light scattering measurement of LP-HER-WOR-MOF (a) and TEM of LP–HER-WOR-MOF (b).



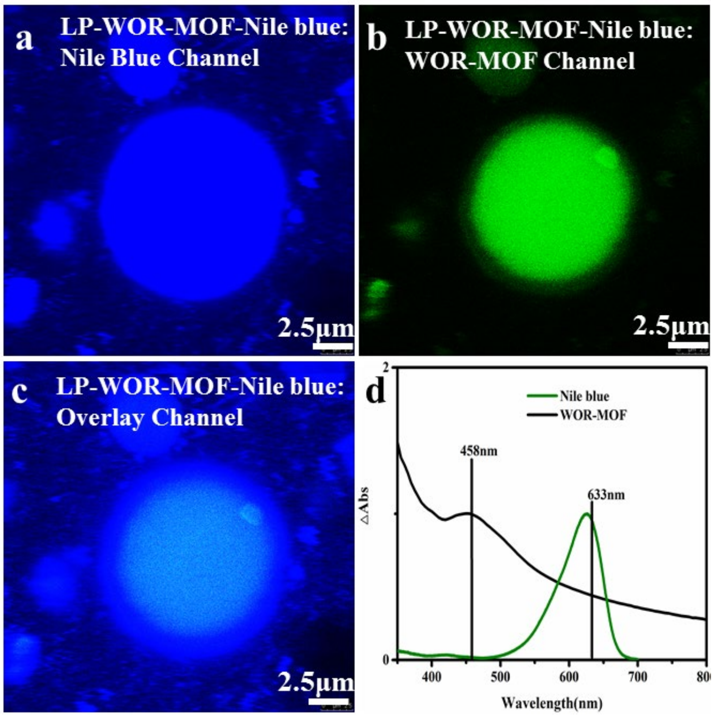
Supplementary Figure 24. TEM images of the same LP-HER-WOR-MOF. Demonstrate the heterogeneity of the system.



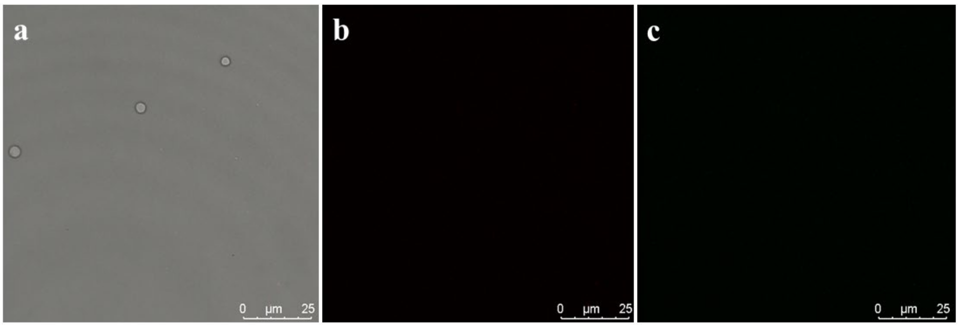
Supplementary Figure 25. Confocal fluorescence microscopy images of LP-HER-MOF. The emission from [(TCPP)Zn] in the HER-MOF (excited at 405 nm, emission in the range of 600-700nm) can be observed either all over the liposome or only on the ring (a)&(b).



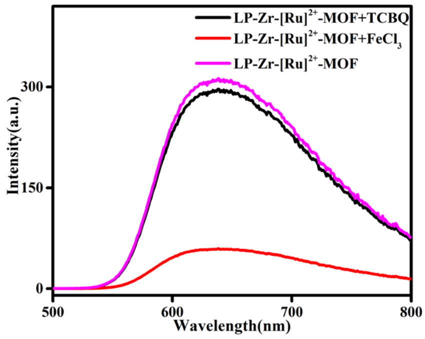
Supplementary Figure 26. Confocal fluorescence microscopy images of LP-HER-MOF-Nile blue in different channels. Note that the LP-HER-MOF in this figure contains 100% [(TCPP)Pt] for its spectrum seperation from Nile blue. (a) is the HER-MOF channel with excitation wavelength of 405 nm and emission wavelength in the range of 600-700nm. (b) is the Nile blue channel with excitation wavelength of 633nm and emission wavelength in the range of 600-700nm. (c) is the overlay channel. (d) is the UV-Vis spectrum. There is no excitation spectral overlap between the two. There is possile energy tranfer between the two, but because they are differentiated by the excitation wavelength, the energy transfer will not cause missassignment.



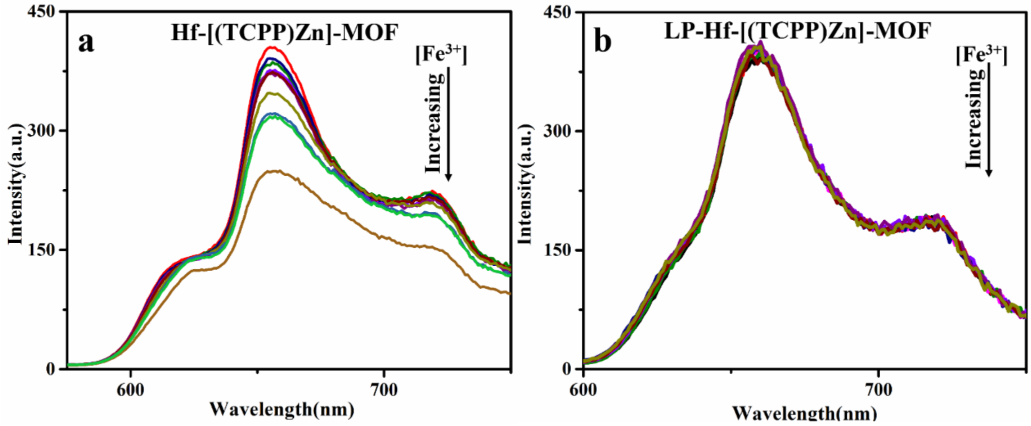
Supplementary Figure 27. Confocal fluorescence microscopy images of LP-WOR-MOF-Nile blue in different channels. Note that the LP-WOR-MOF used in this figure contains only $[Ru]^{2+}$ without the Ir centers as the latter can quench the phosphorescence. (a) is the Nile blue channel with excitation wavelength of 633nm and emission wavelength in the range of 600-700nm. (b) is the WOR-MOF channel with excitation wavelength of 458 nm and emission wavelength in the range of 600-700nm. (c) is the overlay channel. (d) is the UV-Vis spectrum. There is no exciation spectral overlap between the two. There is possile energy tranfer between the two, but because they are differentiated by the excitation wavelength, the energy transfer will not cause missassignment.



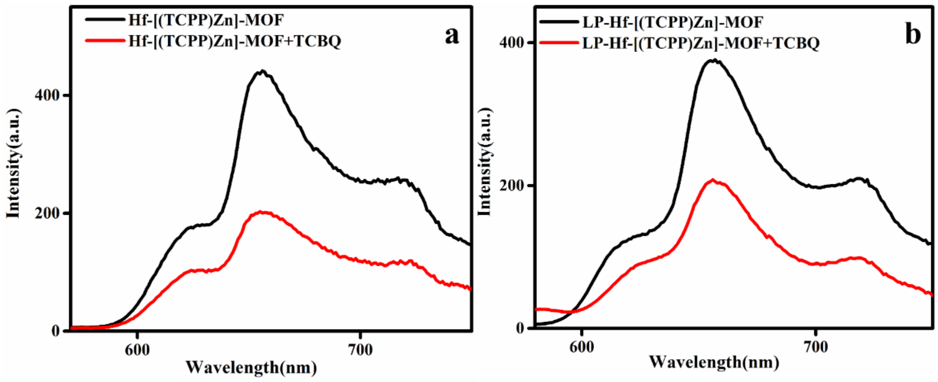
Supplementary Figure 28. Confocal fluorescence microscopy images of blank liposome in different channels. (a) brightfield picture. (b) excitation wavelength was 488nm, emission was collected in the range of 600-700 nm. (c) excitation wavelength was 552nm, emission was collected in the range of 600-700 nm.



Supplementary Figure 29. Fluorescence emission spectra of LP-Zr-[Ru]$^{2+}$-MOF, LP-Zr-[Ru]$^{2+}$-MOF+TCBQ and LP-Zr-[Ru]$^{2+}$-MOF+Fe^{3+}. The fact that the emission of the LP-Zr-[Ru]$^{2+}$-MOF can be quenced by water soluble Fe^{3+} but cannot be efficiently quenced by lipid soluble TCBQ indicates that the Zr-[Ru]$^{2+}$-MOF is in the aqueous phase.



Suppelementary Figure 30. Fluorescence quenching by adding $FeCl_3$ to the aqueous suspension of Hf-[(TCPP)Zn]-MOF (a) and to the liposome hybrid LP-Hf-[(TCPP)Zn]-MOF (b).The concentration of the quencher is from 100μM to 650 μM increased at 50 μM intervals. We find that Fe^{3+}can not quench the emission of the LP-Hf-[(TCPP)Zn]-MOF, which shows effective seperation between Fe^{3+} (in aqueous phase) and the Hf-[(TCPP)Zn]-MOF (in the lipid bilayer) by the liposome structure.



Supplementary Figure 31. Fluorescence quenching by adding TCBQ to the Hf-[(TCPP)Zn]-MOF (a) and to the liposome hybrid LP-Hf-[(TCPP)Zn]-MOF (b).The concentration of quencher is 1 mM. We find that TCBQ can quench the emission of the LP-Hf-[(TCPP)Zn]-MOF, which shows TCBQ (in the lipid bilayer) and the Hf-[(TCPP)Zn]-MOF are in the same position at the liposome structure.

6. Procedures and Results of HER

Supplementary Table 1. Elemental analysis of HER-MOF and LP-HER-MOF[a]

Sample	Pt/(Zn+Pt) (molar ratio)	Sample	Pt/(Zn+Pt) (molar ratio)	Sample	Pt/(Zn+Pt) (molar ratio)
HER-MOF-1	0.034%	HER-MOF-11	0.79%	LP-HER-MOF-1	0.013%
HER-MOF-2	0.086%	HER-MOF-12	0.87%	LP-HER-MOF-2	0.098%
HER-MOF-3	0.129%	HER-MOF-13	1.02%	LP-HER-MOF-3	0.108%
HER-MOF-4	0.134%	HER-MOF-14	1.13%	LP-HER-MOF-4	0.22%
HER-MOF-5	0.149%	HER-MOF-15	1.86%	LP-HER-MOF-5	0.49%
HER-MOF-6	0.203%	HER-MOF-16	2.56%	LP-HER-MOF-6	0.84%
HER-MOF-7	0.213%	MOF-17	5.18%	LP-HER-MOF-7	8.0%
HER-MOF-8	0.355%	MOF-18	5.76%	LP-HER-MOF-8	0.47%
HER-MOF-9	0.371%				
HER-MOF-10	0.49%				

[a]The doping level of the HER-MOF, which is defined as the ratio of C[(TCPP)Pt] /(C[(TCPP)Pt]+C[(TCPP)Zn]) as determined by ICP-MS analysis of Pt and Zn.

Supplementary Table 2. Summarizing recent reports of photocatalytic hydrogen evolution

Entry	Catalyst	Electronic sacrificial agent	TON	References
1	HER-MOF with [(TCPP)Pt] center	Phenol	290000(24h)	In this work
2	Pt	TEA[a]	7000	J. Am. Chem. Soc. 2012, 134,7211 – 7214[5]
3	TBA-$P_2W_{18}Ni_4$[b]	TEOA[c]	6500	J. Am. Chem. Soc. 2014, 136,14015 – 14018[6]
4	Pt	DMA[d]	343	J. Am. Chem. Soc. 2016, 138, 8698-8701[7]
5	Pt	SA[e]	2011.8(5h)	Angew. Chem. Int. Ed. 2018, 57, 12106 –12110[8]
6	Co	AA[f]	9000	Dalton Trans., 2012,41, 13004-13021[9]
7	Pt	EDTA[g]	135	J. Phys. Chem. B, 2006, 110 (30), 14792–14799[10]

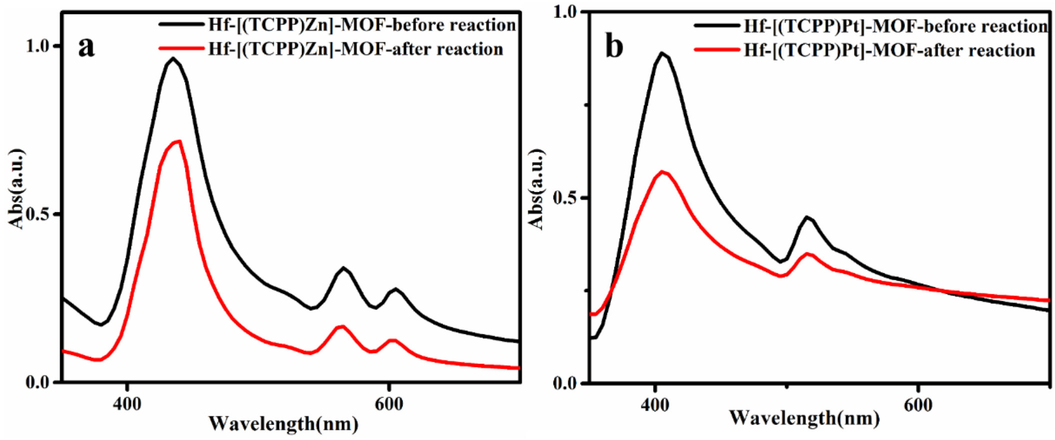
[a]TEA = Triethylamine, [b]TBA= tetrabutylammonium, [c]TEOA= Triethanolamine, [d]DMA = Dimethylacetamide, [e]SA = Sodium ascorbate, [f]AA = Ascorbic acid/Sodium ascorbate, [g]EDTA = Ethylenediaminetetraacetic acid.

Supplementary Table 3. Control experiments for photocatalytic hydrogen evolution[a]

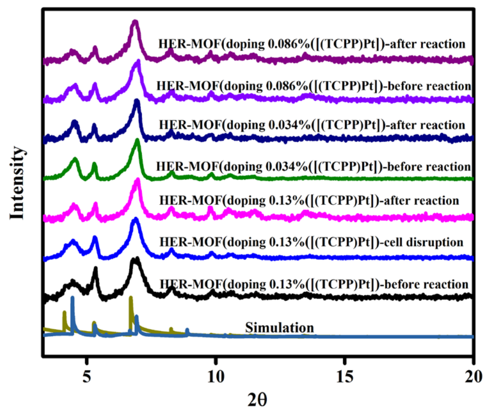
Entry	Catalyst	Electronic sacrificial agent	Pt-TON[b]
1	HER-MOF[c]	No	no H_2 detected
2	HER-MOF[d]	phenol	no H_2 detected
3	Hf-[(TCPP)Zn]-MOF[e]	phenol	no H_2 detected
4	Hf-[(TCPP)Pt]-MOF[f]	phenol	5
5	LP-HER-MOF[g]	No	no H_2 detected
6	LP-HER-MOF[d]	phenol	no H_2 detected
7	LP- Hf-[(TCPP)Zn]-MOF[e]	phenol	no H_2 detected
8	LP- Hf-[(TCPP)Pt]-MOF[f]	phenol	8
9	[(TCPP)Pt][h]	Cholesterol	trace (<0.1)
10	[(TCPP)Pt][h]	Lecithin (PC)	trace (<0.1)
11	[(TCPP)Pt][h]	Cholesterol + TCBQ	trace (<0.1)
12	[(TCPP)Pt][h]	Lecithin (PC) + TCBQ	no H_2 detected
13	LP-HER-MOF[i]	Fe^{3+} (1.4 eq.) and TCBQ (1.4 eq.)	trace (<0.1)
14	LP-HER-MOF[i]	TCBQ (1.4 eq.)	no H_2 detected
15	LP-HER-MOF[i]	Fe^{3+} (1.4 eq.)	no H_2 detected
16	LP-HER-WOR-MOF (12h)	Fe^{3+} (1.4 eq.) and TCBQ (1.4 eq.)	**47**
17	LP-HER-WOR-MOF (72h)	Fe^{3+} (1.4 eq.) and TCBQ (1.4 eq.)	223

[a]Hydrogen evolution reactions were carried out in water for 24 hours using a 400 nm LED. [b] Pt-based turnover number (Pt-TON) is defined as $n(H_2)/n(Pt)$. [c]without adding phenol. [d]in dark. [e]without doping the [(TCPP)Pt] into the MOF. [f]with 100% of [(TCPP)Pt] in the MOF. [g]without adding phenol. [h]in THF/H_2O =9/1 solution,

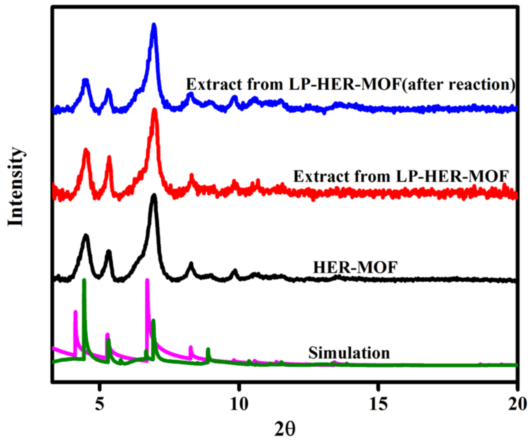
[(TCPP)Pt] 0.1 mg/mL. Concentrations of sacrificial reagents: phenol (110 mM for entries 2-4, 0.5 $mg/mL_{liposome}$ for entries 6-8), Cholesterol (110 mM), PC (110 mM), TCBQ (0.4 mM). [i]The amounts of HER-MOF and both shuttles were the same as those in the water splitting tests. Concentrations of sacrificial reagents: TCBQ (0.018 mM for 13&14), Fe3+ (0.018 mM for 13&15), the amount of HER-MOF is 0.07 mg/mL.



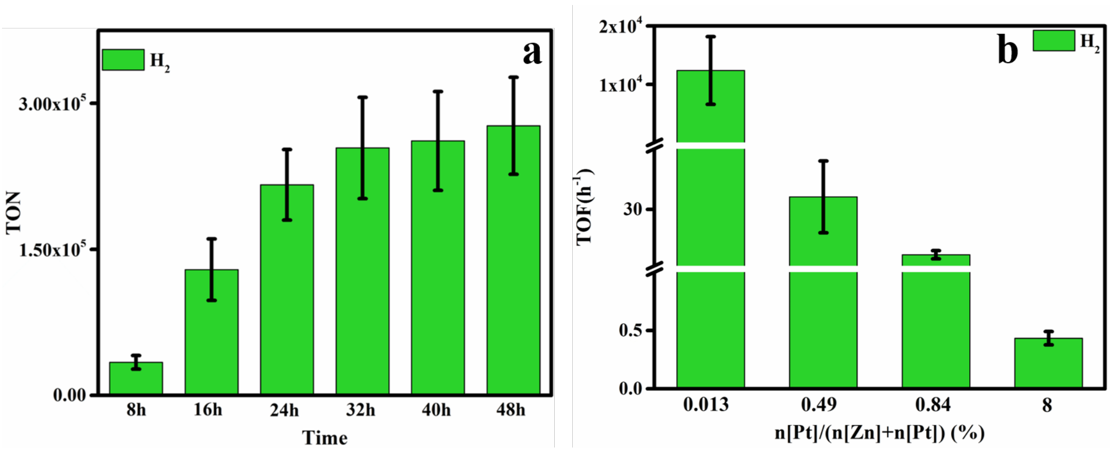
Supplementary Figure 32. UV-Vis spectra of Hf-[(TCPP)Zn]-MOF (a) and UV-Vis spectra Hf-[(TCPP)Pt]-MOF (b) before and after reaction.The two peaks of the Q-bands indicate that no Zn and Pt was removed from the porphyrin ring after the catalysis, as the TCPP with two hydrogen atoms at the center of the ring will show two additional absorption peaks in the UV-Vis spectra.



Supplementary Figure 33. PXRD patterns of HER-MOF before and after reaction.



Supplementary Figure 34. PXRD patterns of HER-MOF (or LP-HER-MOF) before and after assembling with liposome and after reaction.



Supplementary Figure 35. The photocatalytic hydrogen evolution by the LP-HER-MOF with different [(TCPP)Pt] doping levels under visible-light irradiation, light source: 400 nm LED. In LP-HER-MOF, the amount of HER-MOF is 0.07mg/mL, the concentration of phenol 0.5 mg/$mL_{liposome}$. The error bar represents ± standard deviation.

7. Procedures and Results of WOR

Supplementary Table 4. Summarizing the recent report of photocatalytic water oxidation

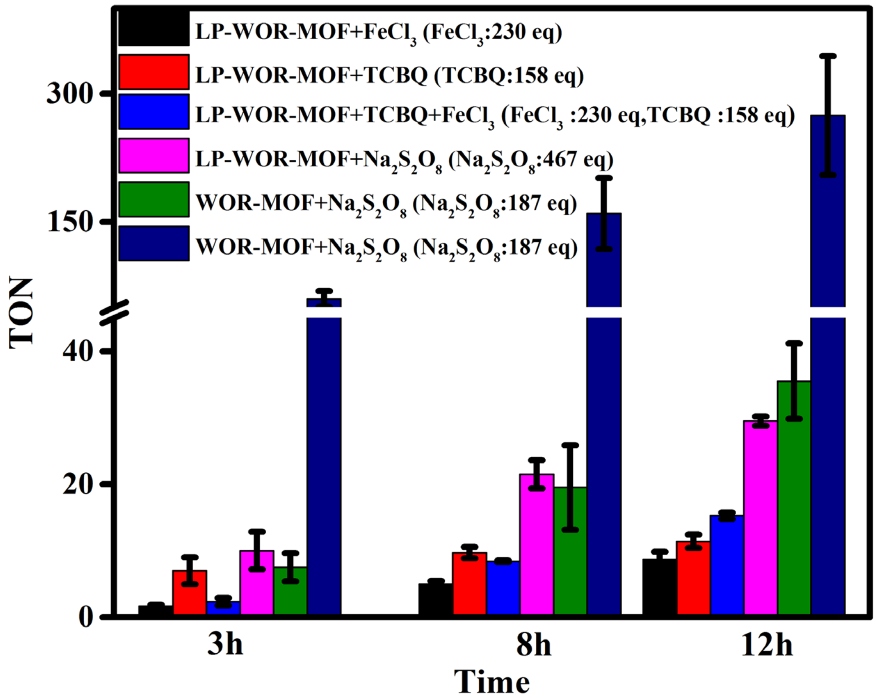
Entry	Catalyst	Electron sacrificial agent	TON[a]	References
1	WOR-MOF with [(BPYDC)Ir] center	$Na_2S_2O_8$	252	In this work[b]
2	WOR-MOF with [(BPYDC)Ir] center	$Na_2S_2O_8$	36	In this work[c]
3	POM@Co_3O_4	$Na_2S_2O_8$	0.3 (5 min)	Chem. Eur. J. 2016, 22, 15513 – 15520[11]
4	POM	$Na_2S_2O_8$	36 (15 min)	J. Am. Chem. Soc. 2018, 140, 3613-3618[12]
5	POM	$Na_2S_2O_8$	224	J. Am. Chem. Soc. 2011, 133, 2068-2071[13]

[a]The TON is defined as $n(O_2)/n(M)$ [b]The WOR-MOF here was prepared using $Ir(NO_3)_3{\cdot}3H_2O$. [c]The WOR-MOF here was prepared using $IrCl_3{\cdot}3H_2O$.

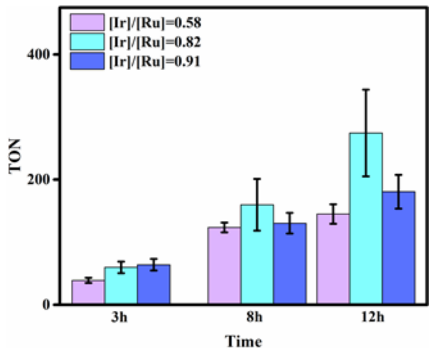
Supplementary Table 5. Control experiments for photocatalytic water oxidation[a]

Entry	Catalyst	Electron sacrificial agent	Ir-TON[b]
1	WOR-MOF [c]	No	no O_2 detected
2	Zr-$[Ru]^{2+}$-MOF[d]	$Na_2S_2O_8$	no O_2 detected
3	WOR-MOF[e]	$Na_2S_2O_8$	no O_2 detected
4	LP-WOR-MOF[c]	No	no O_2 detected
5	LP-Zr-$[Ru]^{2+}$-MOF [d]	$Na_2S_2O_8$	no O_2 detected
6	LP-WOR-MOF[e]	$Na_2S_2O_8$	no O_2 detected
7	LP-Zr-$[Ru]^{2+}$-MOF [d]	TCBQ	no O_2 detected
8	LP-WOR-MOF	Fe^{3+} (1.4 eq.) and TCBQ (1.4 eq.)	0.26
9	LP-WOR-MOF	TCBQ (1.4 eq.)	0.15
10	LP-WOR-MOF	Fe^{3+} (1.4 eq.)	0.06
11	LP-HER-WOR-MOF (12h)	Fe^{3+} (1.4 eq.) and TCBQ (1.4 eq.)	2.8
12	LP-HER-WOR-MOF (72h)	Fe^{3+} (1.4 eq.) and TCBQ (1.4 eq.)	14

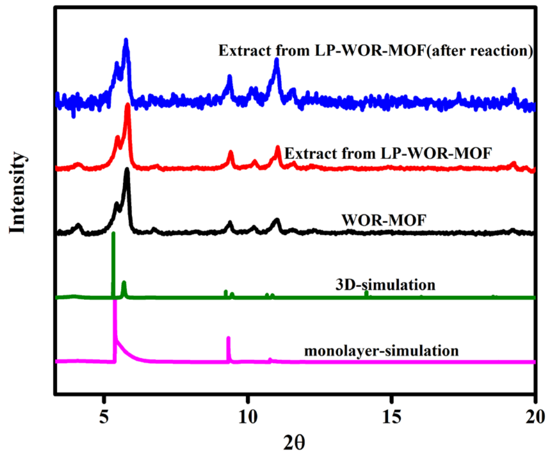
[a] Water oxidation reaction were carried out for 12 hours using a 450 nm LED. [b] Ir-based turnover number (Ir-TON) is defined as $n(O_2)/n(Ir)$. [c]without adding any electron sacrificial agent. [d]without loading the Ir into the MOF. [e] in the dark. Concentrations of sacrificial reagents: $Na_2S_2O_8$ (8 mM for entries 2&3, 4 mM for entries 5&6), TCBQ (2 mM). [f] the amounts of WOR-MOF and both shuttles were the same as those in the water splitting tests. Concentrations of sacrificial reagents : TCBQ (0.018 mM for 8&9), Fe^{3+} (0.018 mM for 8&10), the amount of WOR-MOF is 0.04 mg/mL.



Supplementary Figure 36. Photocatalytic water oxidation under visible-light irradiation using WOR-MOF and LP-WOR-MOF with 450 nm LED irradiation. The WOR-MOF in LP-WOR-MOF was prepared using $IrCl_3$ for Ir functionalization, while some of the WOR-MOF without liposome were functionalized with $Ir(NO_3)_3$. The amount of O_2 was quantified by a fluorescence oxygen sensor. In LP-WOR-MOF, the amount of WOR-MOF is 0.04mg/mL, the amount of $FeCl_3$, TCBQ and $Na_2S_2O_8$ are 0.8mg/mL, 0.5mg/mL and 2mg/mL respectively. In WOR-MOF+$Na_2S_2O_8$, the amount of WOR-MOF is 0.5mg/mL, the amount of $Na_2S_2O_8$ is 10mg/mL. We also planned to do the experiment using TCBQ as oxidant and only WOR-MOF as the catalyst, but TCBQ is insoluble in water, which gave us a lot of difficulty. Attempts to perform this experiment in organic solvent gave inconsistent results, possibly due to competing oxidation of the organic solvents. The error bars represent standard deviations.



Supplementary Figure 37. The photocatalytic water oxidation by the LP-MOF-WOR under visible-light irradiation, light source: 450 nm LED. In LP-WOR-MOF, the amount of WOR-MOF is 0.04mg/mL, the amount of $Na_2S_2O_8$ is 2mg/mL and the WOR-MOFs were functionalized with $Ir(NO_3)_3$. From the figure, we can find that [Ir]/[Ru]=0.82, water oxidation has the best catalytic effect. The error bar represents ± standard deviation.



Supplementary Figure 38. PXRD patterns of WOR-MOF (or LP-WOR-MOF) before and after being assembled into liposomes and after reaction.

Supplementary Figure 39. Experimental device for detecting O_2 using a fluorescence detector. Note that only data in Supplementary Fig. 36 were obtained using the fluorescence detector, while other data on O_2 quantification throughout the paper were obtained by GC using the experimental set-up in Suppementarty Fig. 41.

8. Procedures and Results of Total Water Splitting

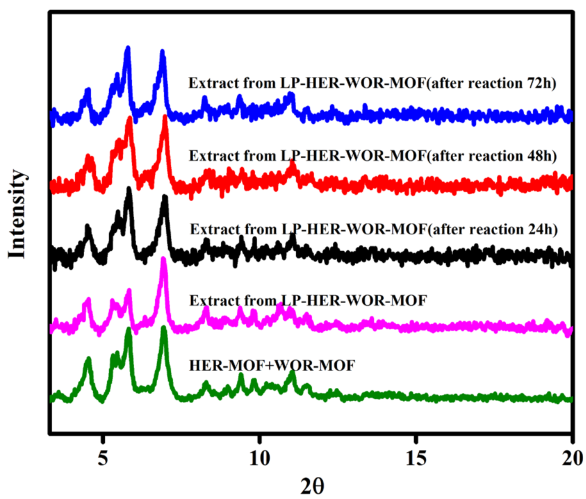
Supplementary Table 6. Summarizing recent reports of photocatalytic total water splitting

Entry	Photocatalyst	Activity	AQE(H_2)	References
1	LP-HER-WOR-MOF	H_2:836 μmol/g O_2:464 μmol/g	(1.5±1) %	In this work
2	Ru/$SrTiO_3$:Rh/$BiVO_4$	H_2:367 μmol/g O_2:183 μmol/g	1.03%	J. Am. Chem. Soc., 2011, 133 (29), 11054–11057[14]
3	Ru/$SrTiO_3$:Rh/$BiVO_4$	H_2:2378 μmol/g O_2:1100 μmol/g	1.29%	Nano Lett. 2018, 18, 805−810[15]
4	BP/$BiVO_4$	H_2:780 μmol/g O_2:460 μmol/g	0.89%	Angew. Chem. Int. Ed. 2018, 57, 2160 –2164[16]
5	Ru/$SrTiO_3$:La/Rh/ $BiVO_4$:Mo	H_2:54.4 mmol/g O_2:27.2 mmol/g	19%	J. Am. Chem. Soc. 2017,139, 1675-1683[17]

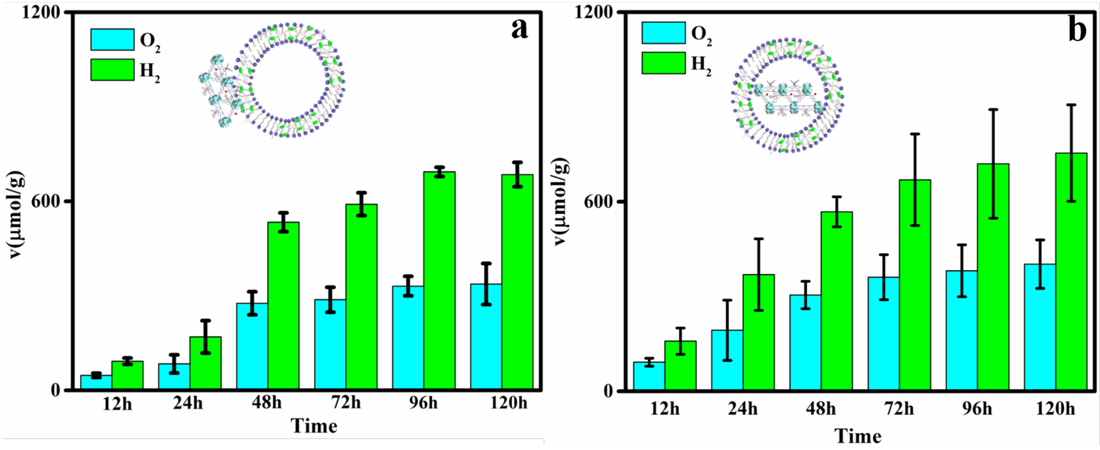
Supplementary Table 7. Control experiments for photocatalytic total water splitting[a]

Entry	Catalyst	Redox relay	activity
1	LP-HER-WOR-MOF[b]	TCBQ	H_2:586 μmol/g O_2:260μmol/g
2	LP-HER-WOR-MOF[c]	$FeCl_3$	H_2:236 μmol/g O_2:163μmol/g
3	HER-WOR-MOF[d]	TCBQ&$FeCl_3$	H_2:99 μmol/g O_2:42μmol/g
4	LP-HER-WOR-MOF[e]	TCBQ&$FeCl_3$	0

[a]photocatalytic total water splitting reactions were carried out for 72 hours using 400nm LED+450nm LED. [b]only using the TCBQ as redox relay. [c]only using the $FeCl_3$ as redox relay. [d]without the liposome, only the HER-MOF and WOR-MOF-Ir in the system along with the redox relays. [e]in dark.

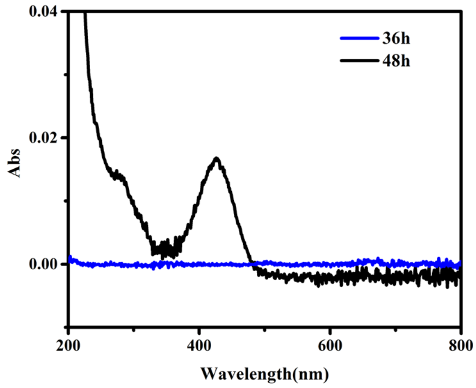


Supplementary Figure 40. PXRD patterns of MOFs extracted from LP-HER-WOR-MOF before and after reaction.

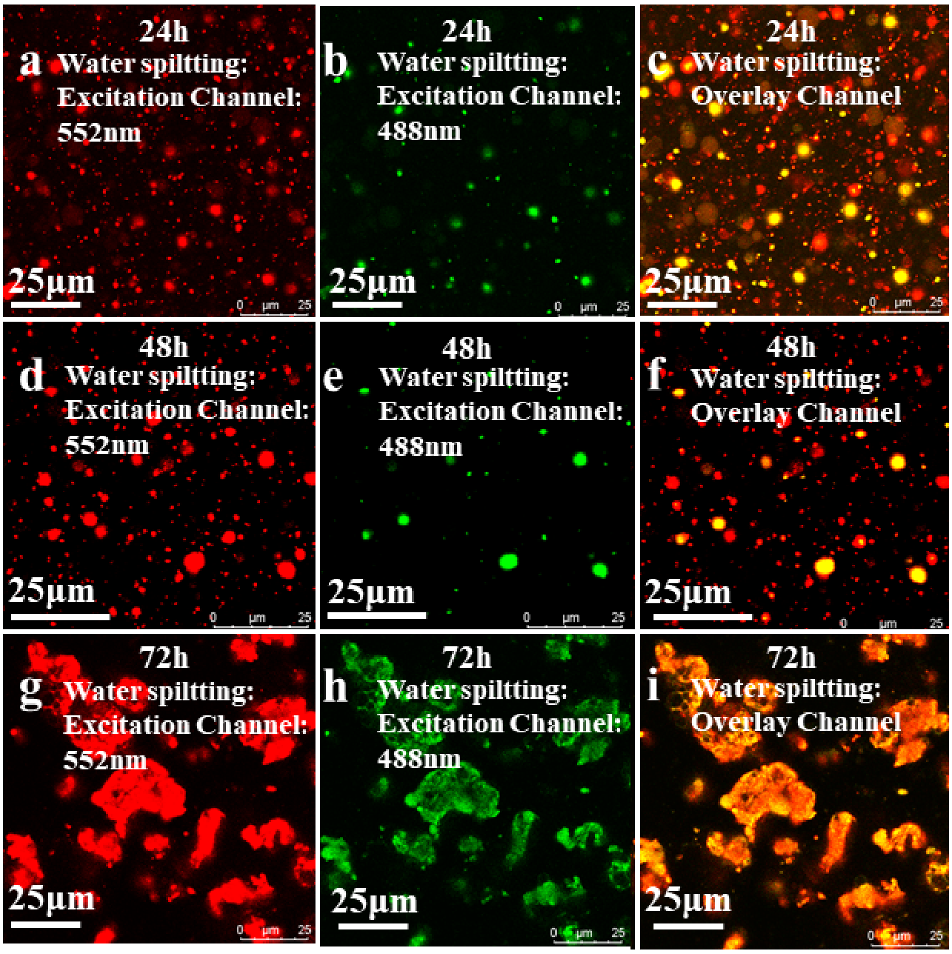


Supplementary Figure 41. (a) The total water splitting by LP-HER-WOR (external water phase)-MOF under visible-light irradiation, light source: 400 nm LED + 450 nm LED. (b) The total water splitting by LP-HER-WOR-MOF under visible-light irradiation, light source: 400 nm LED + 450 nm LED. In (a), the amount of catalysts and shuttles used were the same as that in Figure 2a, and Fe^{3+} was put in the outer water phase like WOR-MOF. The error bar represents ± standard deviation.

Supplementary Figure 42. Experimental setup for total water splitting.

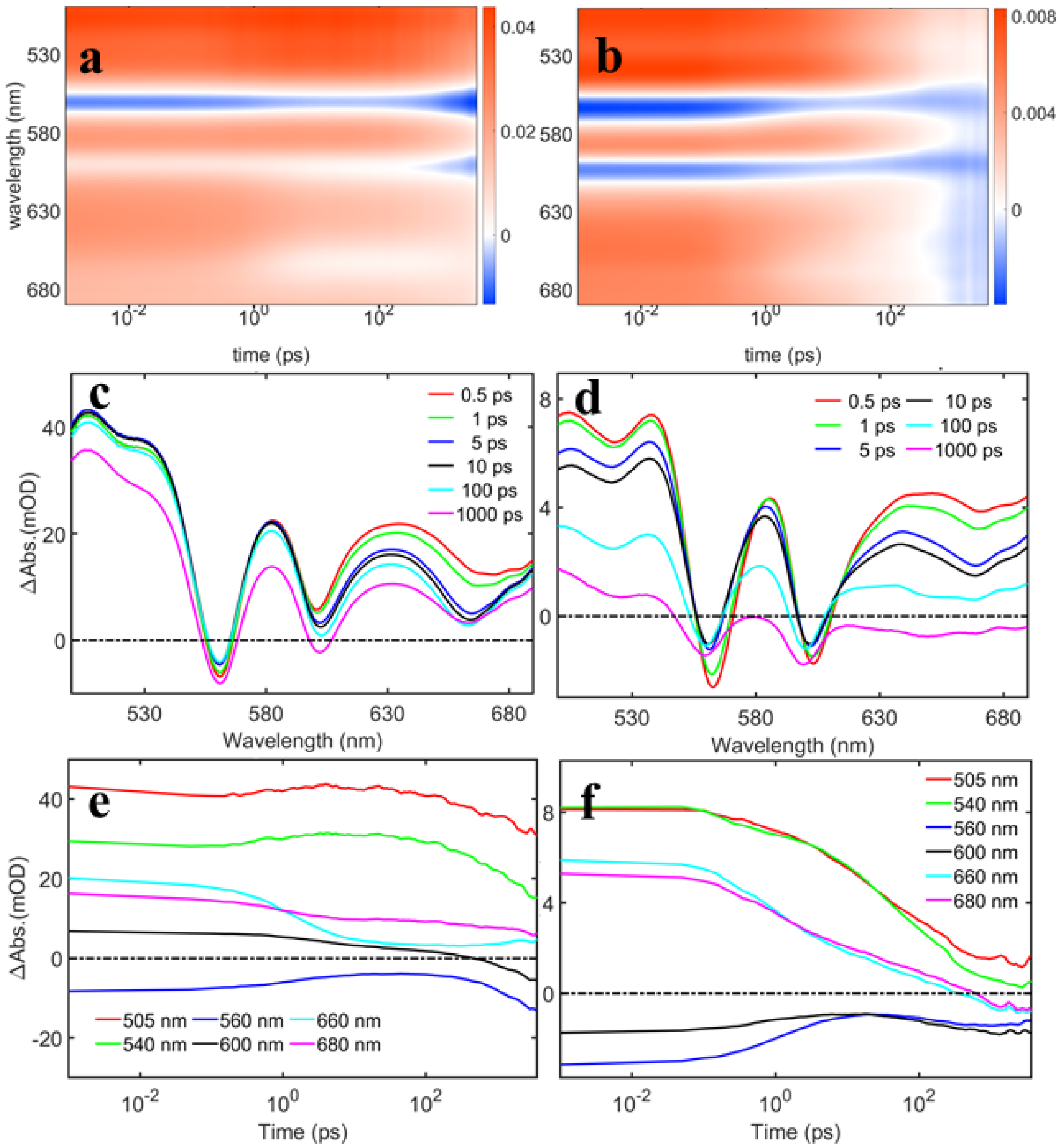


Supplementary Figure 43. UV-Vis spectrum of released coumarin 30. Coumarin 30 was added in the liposomal synthesis and excess Coumarin 30 in the outside solution was separated from the liposomes through dialysis (dialysis membrane = 34 mm, MW =3500 kD). After water splitting reaction of 36 h, we observed no dye release from the liposomes as determined by UV-Vis measurements, but after a reaction of 48 h, we did observe some dye release, which indicates the loss of integrity of the lipid layers after 36 h.

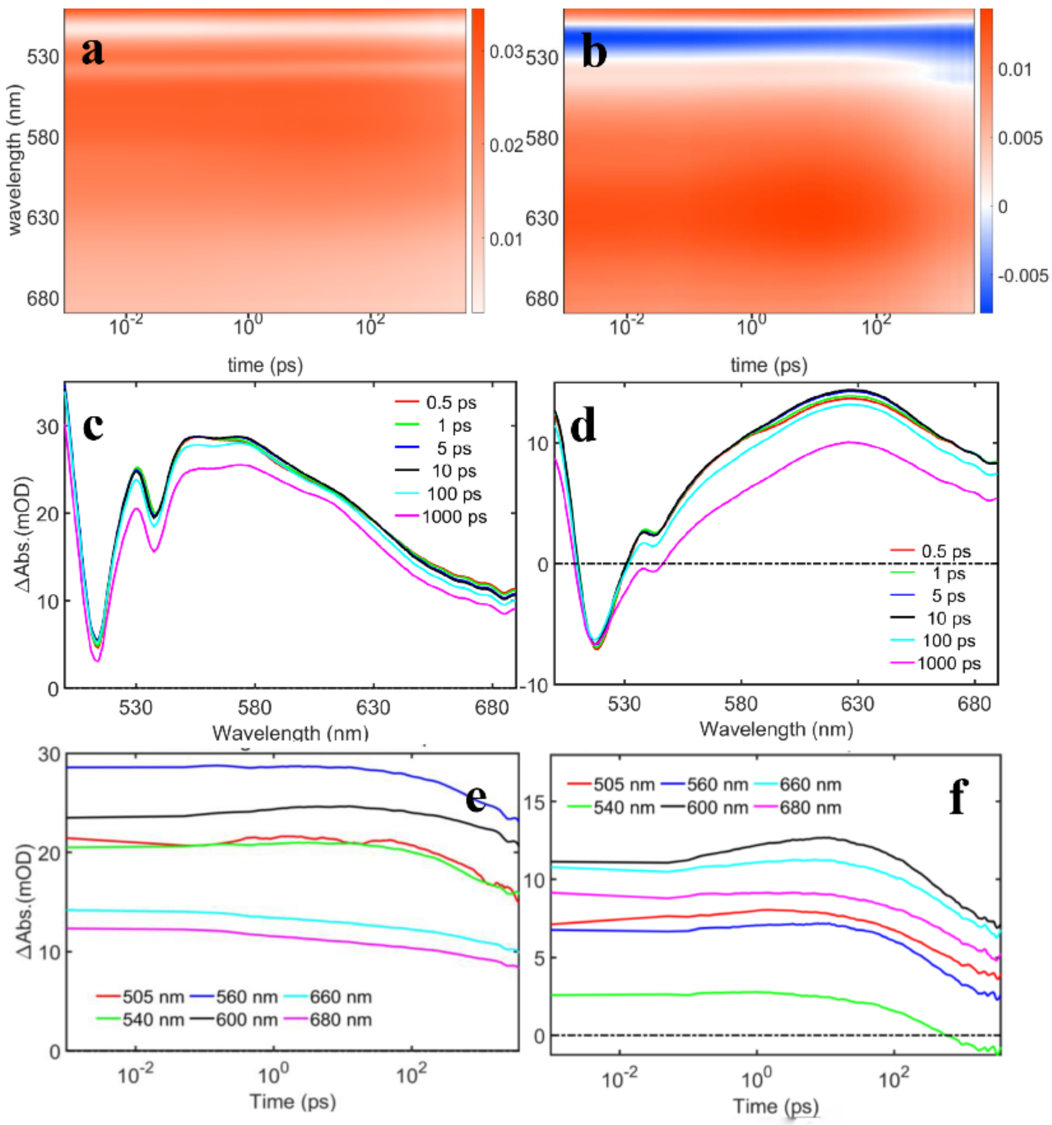


Supplementary Figure 44. Confocal fluorescence microscopy images of LP-HER-WOR-MOF after water splitting reaction. a-c is fluorescence imaging after 24 h water splitting reaction, d-f is fluorescence imaging after 48h water splitting reaction, g-i is fluorescence imaging after 72 h water splitting reaction. All emissions were collected in the wavelength range of 600-700 nm.

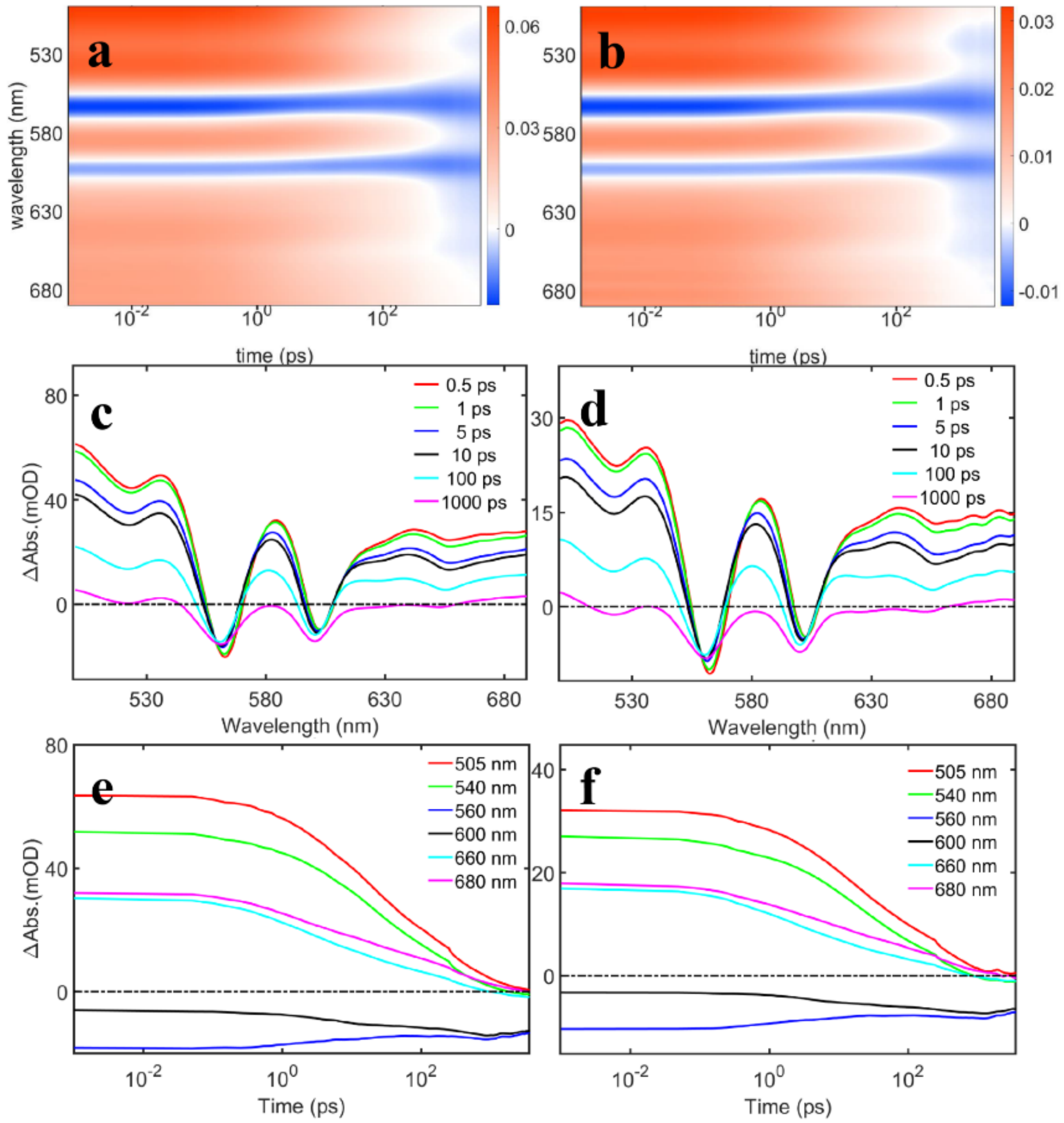
9. Time-Resolved Spectroscopic Analysis of Photocatalytic Process



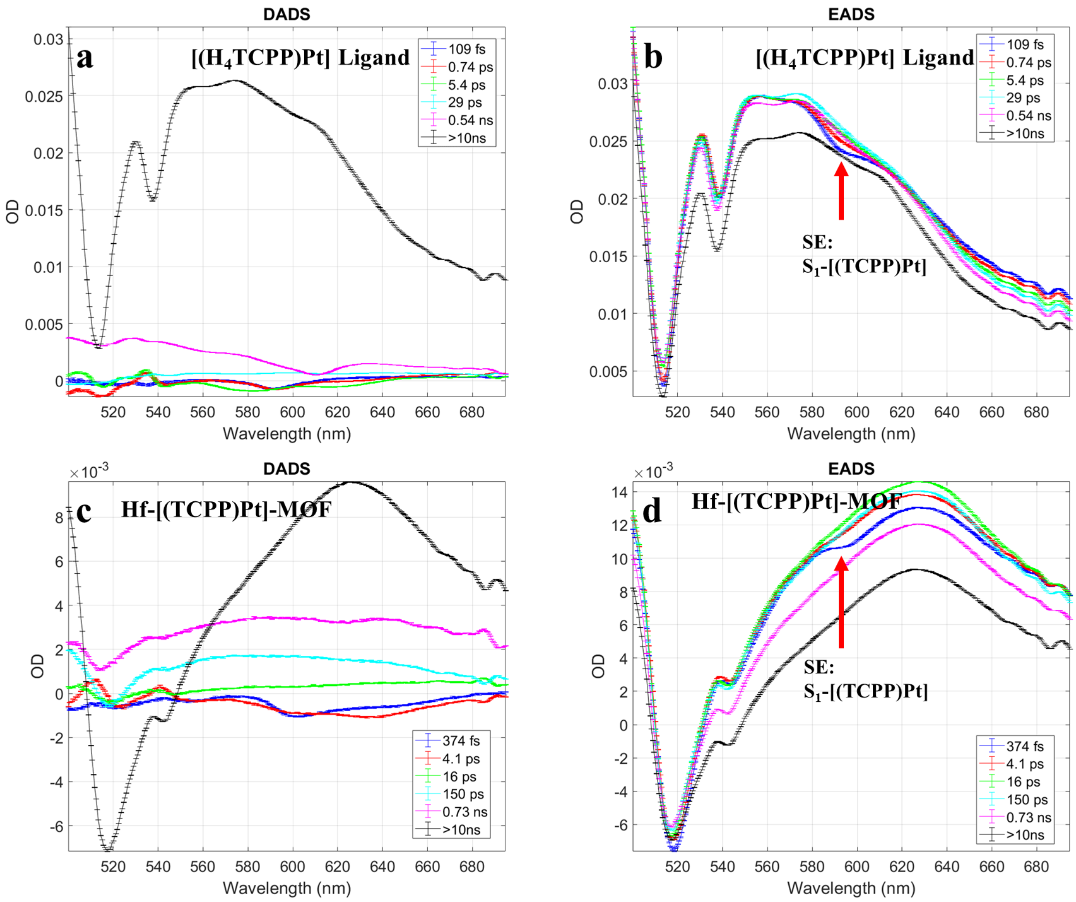
Supplementary Figure 45. Femtosecond transient absorption spectra of [(H_4TCPP)Zn] Ligand in DMF (a, c, e) and Hf-[(TCPP)Zn]-MOF (b, d, f).



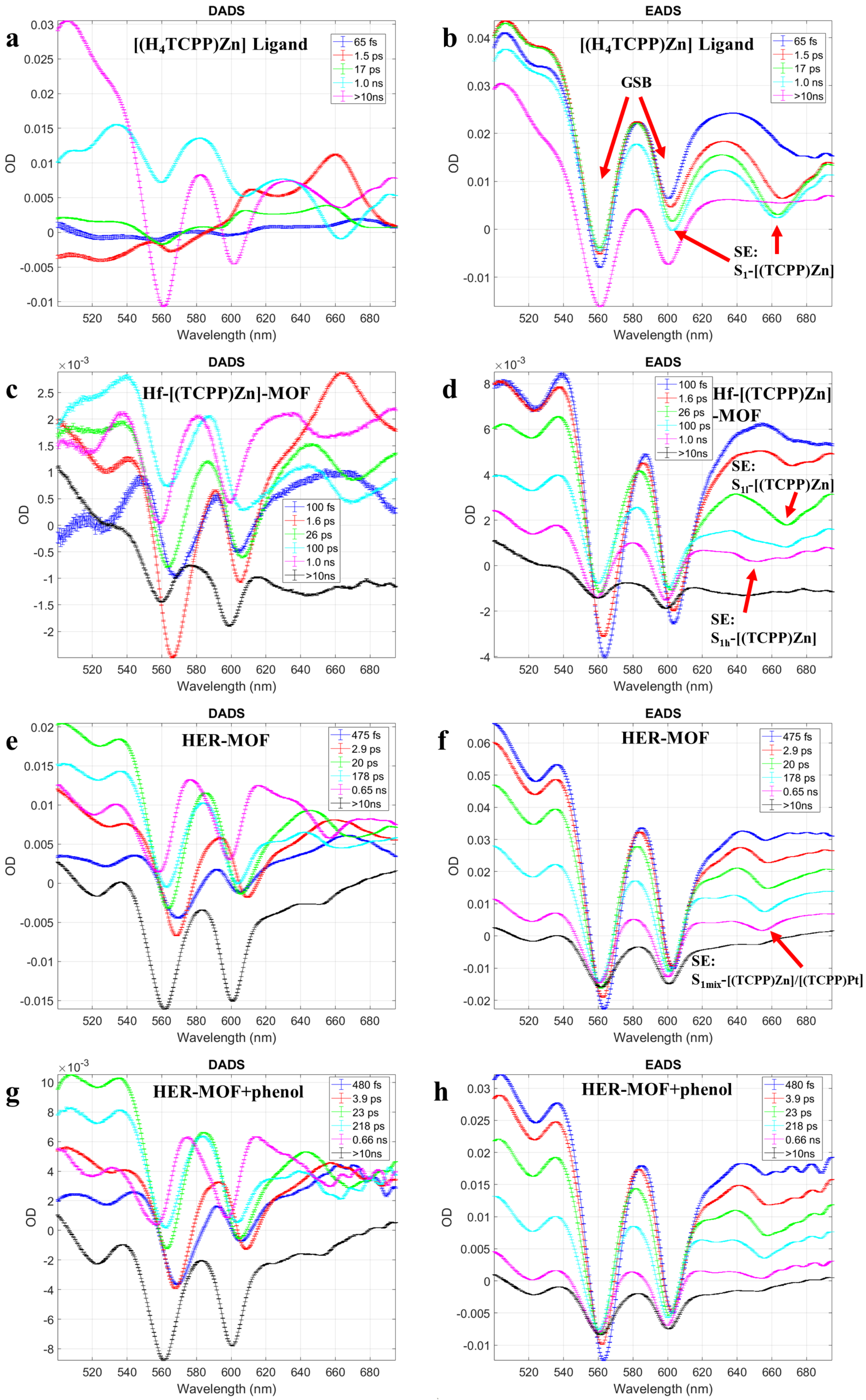
Supplementary Figure 46. Femtosecond transient absorption spectra of [(H_4TCPP)Pt] Ligand (a, c, e) and Hf-[(TCPP)Pt]-MOF (b, d, f).



Supplementary Figure 47. Femtosecond transient absorption spectra of HER-MOF (a, c, e) and HER-MOF + phenol (b, d, f).



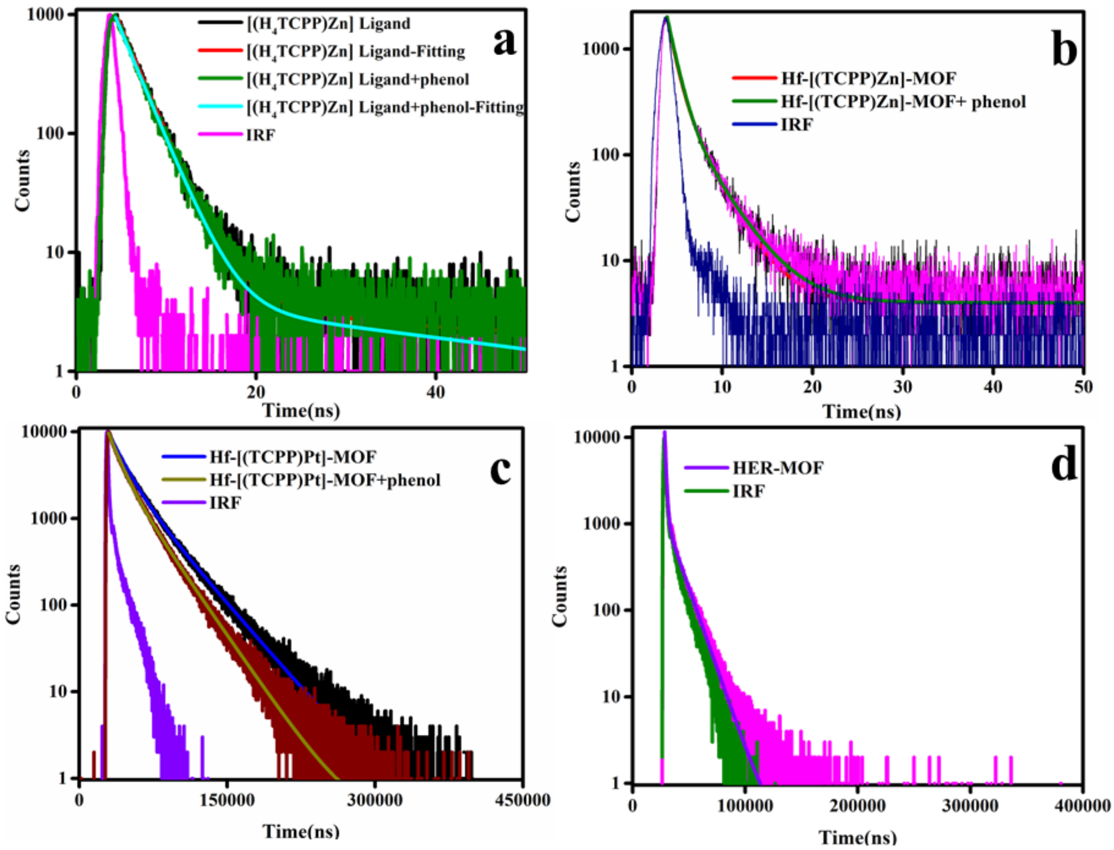
Supplementary Figure 48. The DADS and EADS from global fittings of fsTA spectra of [(H$_4$TCPP)Pt] ligand (a)&(b) and Hf-[(TCPP)Pt]-MOF (c)&(d).



Supplementary Figure 49. The DADS and EADS from global fittings of fsTA spectra of [(H_4TCPP)Zn] ligand (a)&(b), Hf-[(TCPP)Zn]-MOF (c)&(d), HER-MOF (e)&(f), and HER-MOF + phenol (g)&(h).

Supplementary Table 8. Time constants from global fitting of fsTA spectra of Hf-[(TCPP)Zn(Pt)] MOFs and homogeneous ligands

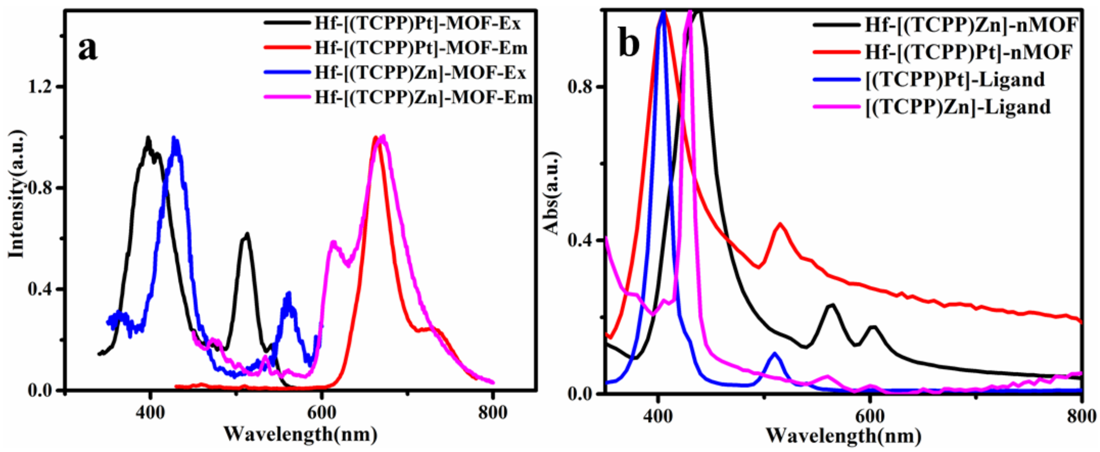
Sample	Time constants					
	1st	2nd	3rd	4th	5th	6^{th}
[(H_4TCPP)Zn] Ligand	65fs	1.5ps	17ps	-	1.0ns	>10ns
Hf-[(TCPP)Zn]-MOF	100fs	1.6ps	26ps	100ps	1.0ns	>10ns
HER-MOF	475fs	2.9ps	20ps	178ps	0.65ns	>10ns
HER-MOF + phenol	480fs	3.9ps	22ps	218ps	0.66ns	>10ns
[(H_4TCPP)Pt] Ligand	109fs	0.74ps	5.4ps	29ps	0.54ns	>10ns
Hf-[(TCPP)Pt]-MOF	374fs	4.1ps	16ps	150ps	0.73ns	>10ns



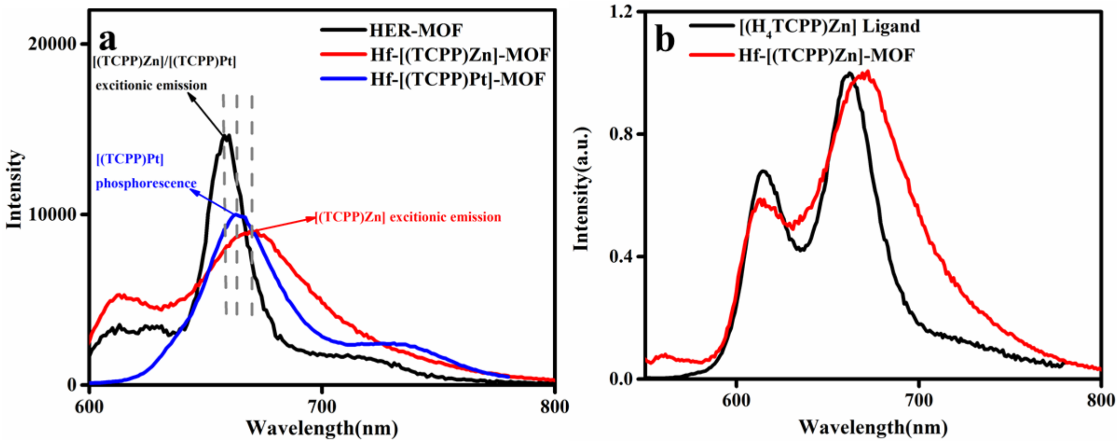
Supplementary Figure 50. Time resolved fluorescence measurements of [(H_4TCPP)Zn] ligand and [(H_4TCPP)Zn] ligand+phenol (a), Hf-[(TCPP)Zn]-MOF and Hf-[(TCPP)Zn]-MOF+phenol (b), Hf-[(TCPP)Pt]-MOF and Hf-[(TCPP)Pt]-MOF+phenol (c), and HER-MOF (d).

Supplementary Table 9. The fitting parameters of the time-resolved fluorescence data in Supplementary Fig. 48.

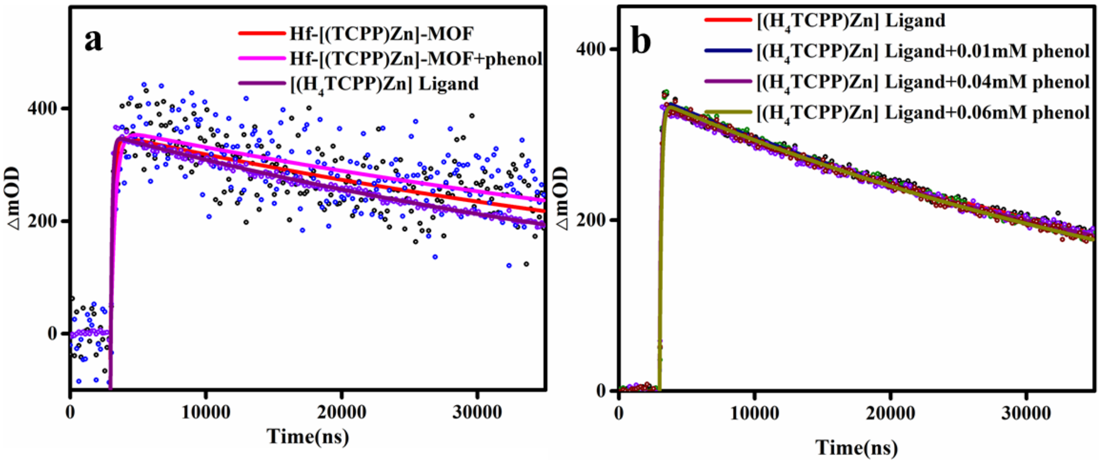
Sample	B1	τ1	B2	τ2	Rel%	Weighted average τ
[(H_4TCPP)Zn] Ligand	941	2.4ns	5	13.4ns	96.90(τ1), 3.10(τ2)	2.5ns
[(H_4TCPP)Zn] Ligand+phenol	960	2.4ns	4	50ns	91.31(τ1), 8.69(τ2)	2.6ns
Hf-[(TCPP)Zn]-MOF	1851	0.95ns	330	3.2ns	62.44(τ1), 37.56(τ2)	1.3ns
Hf-[(TCPP)Zn]-MOF+phenol	1747	0.95ns	311	3.2ns	62.39(τ1), 37.61(τ2)	1.3ns
Hf-[(TCPP)Pt]-MOF	5751	14.4μs	3811	33.36μs	39.43(τ1), 60.57(τ2)	22μs
Hf-[(TCPP)Pt]-MOF+phenol	5405	11.08μs	4282	26.38μs	34.64(τ1), 65.36(τ2)	17.8μs
HER-MOF	10804	1.21μs	1170	11.62μs	49.03(τ1), 50.97(τ2)	2.1μs



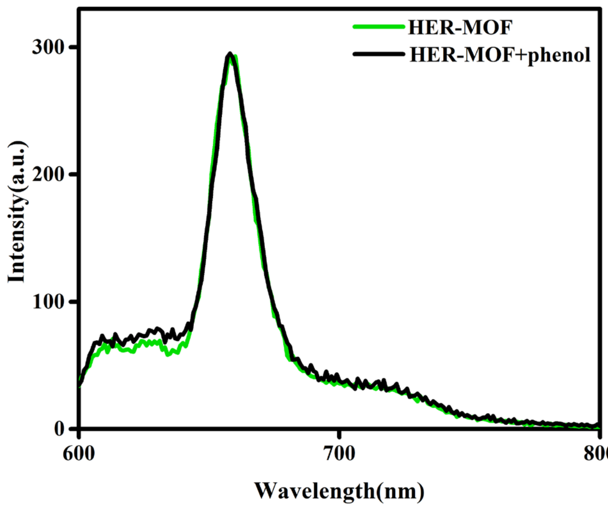
Supplementary Figure 51. The fluorescence emission and excitation spectra of Hf-[(TCPP)Pt]-MOF & Hf-[(TCPP)Zn]-MOF (a), and the UV–vis spectra of Hf-[(TCPP)Pt]-MOF, Hf-[(TCPP)Zn]-MOF, [(TCPP)Pt] ligand and [(TCPP)Zn] ligand (b).



Supplementary Figure 52. Fluorescence emission spectra comparing emissions of HER-MOF, Hf-[(TCPP)Zn]-MOF and Hf-[(TCPP)Pt]-MOF showing different excitonic states in the three (a); Fluorescence emission spectra comparing emissions of the Hf-[(TCPP)Zn]-MOF and solution of the [(H_4TCPP)Zn] ligand to show possible excitonic state in the MOF (b).



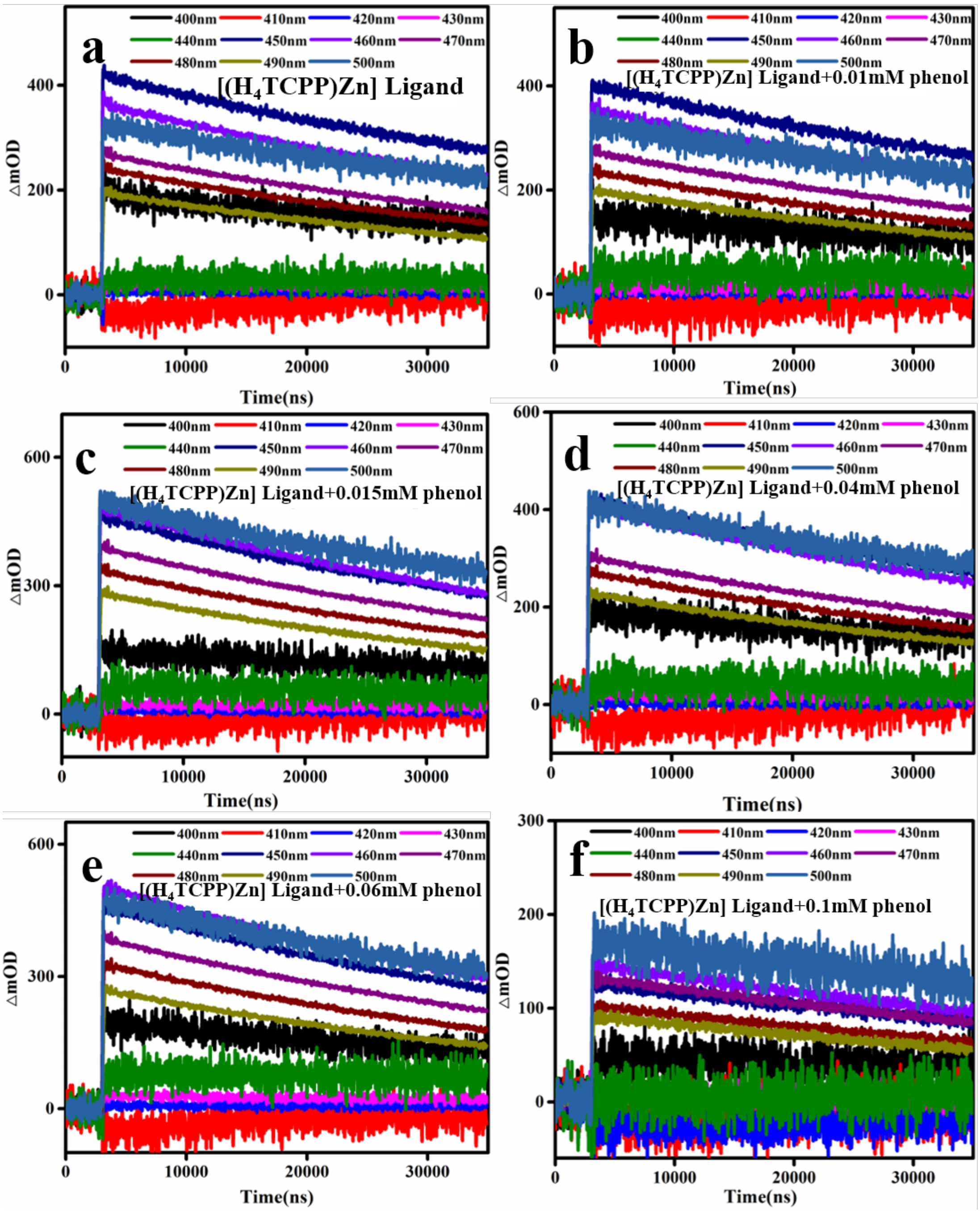
Supplementary Figure 53. The time traces of nanosecond transient absorption spectra (490 nm) of Hf-[(TCPP)Zn]-MOF, Hf-[(TCPP)Zn]-MOF+phenol, solution of [(H_4TCPP)Zn] ligand (a) and [(H_4TCPP)Zn] ligand with different concentrations of phenol (0-0.06 mM) (b). The excitation wavelength is 532 nm.



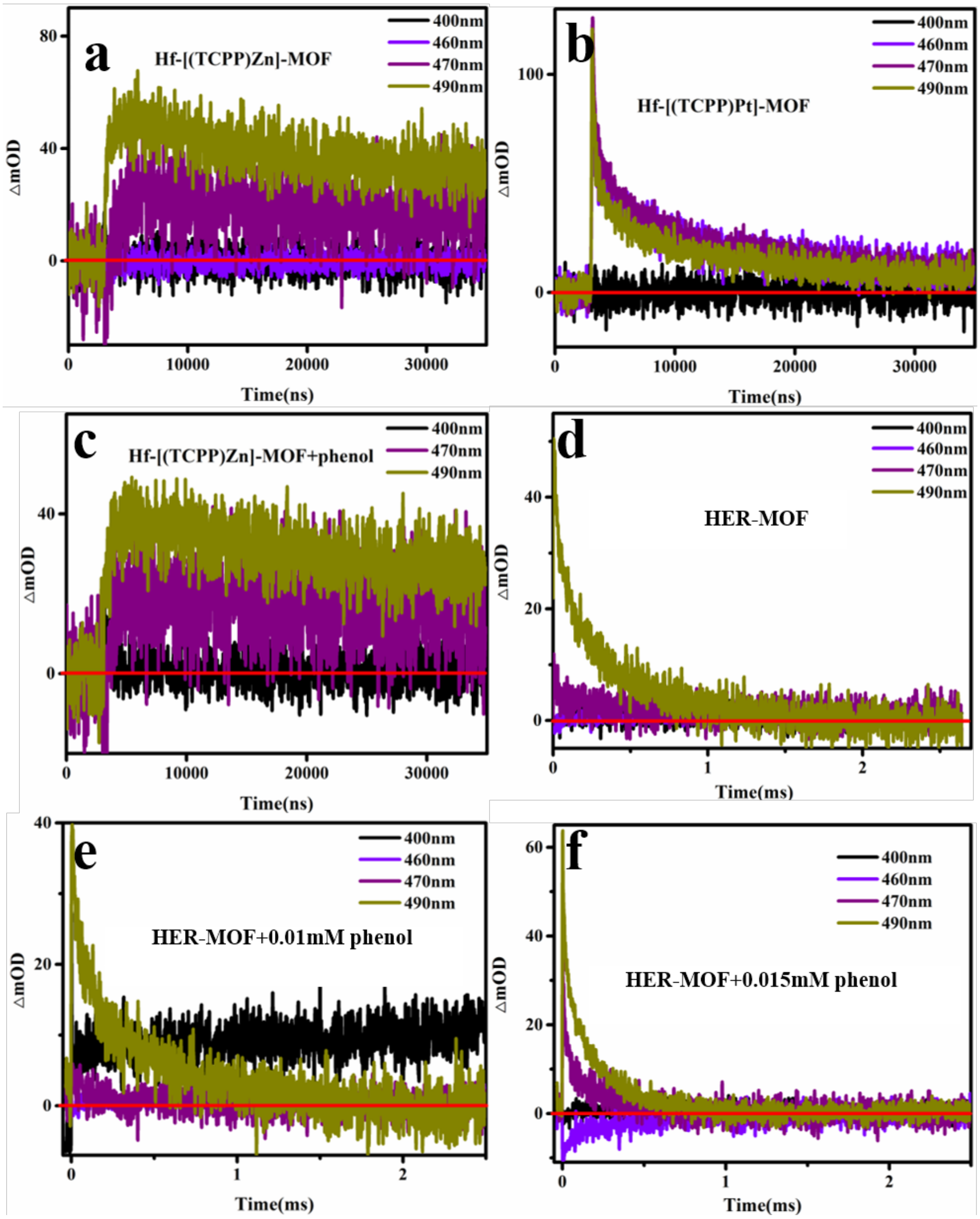
Supplementary Figure 54. Fluorescence emission spectra of HER-MOF (green trace) and HER-MOF+phenol (black trace).

Supplementary Table 10. The fitting parameters of the Supplementary Fig. 51-54**.**

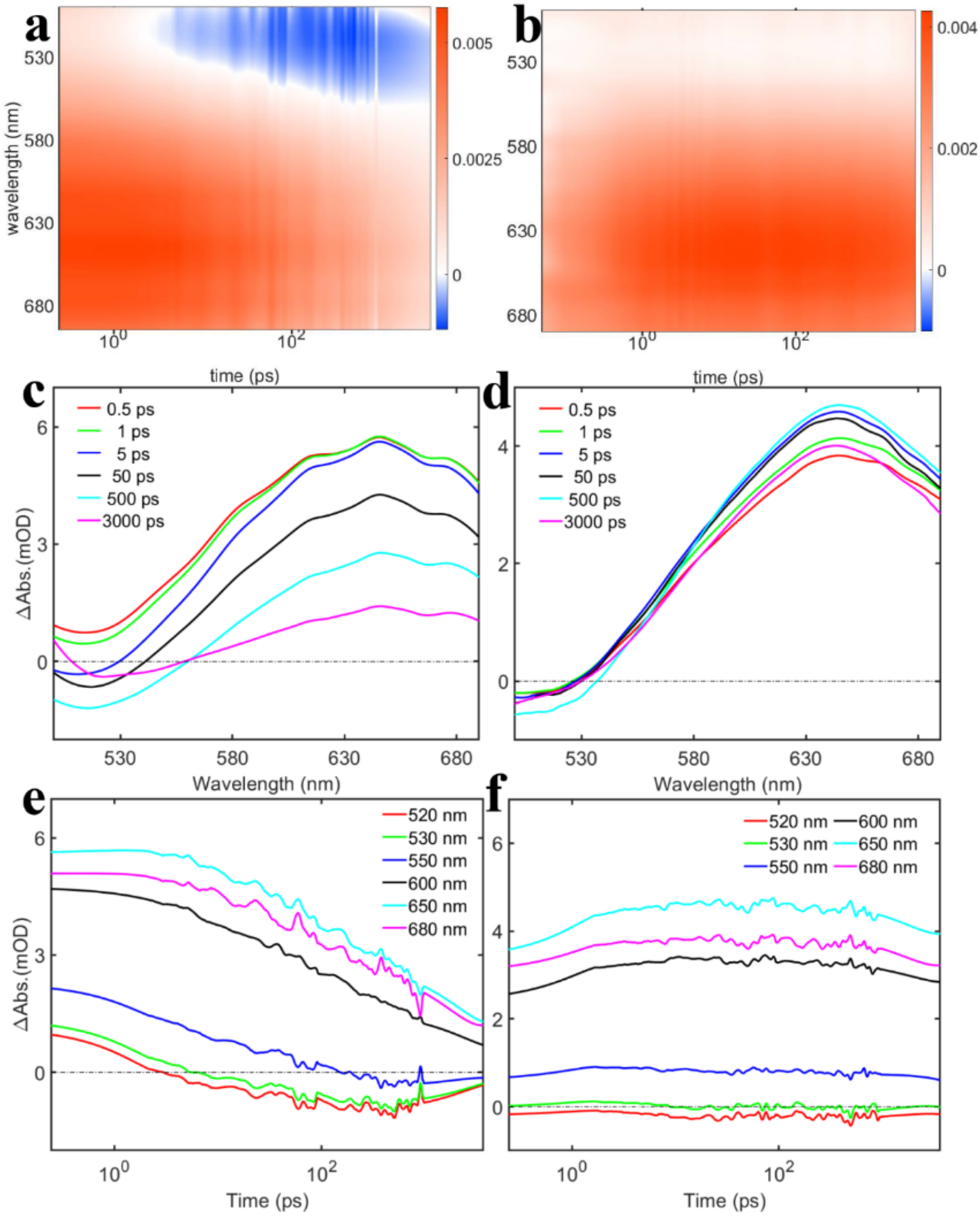
Sample	A1	τ1	A2	τ2	A3	τ3
HER-MOF	33	0.3ms				
HER-MOF+0.01mM phenol	25	0.4ms				
HER-MOF+0.015mM phenol	35	0.2ms				
Hf-[(TCPP)Pt]-MOF	-169	59ns	92	514ns	33	17601ns
Hf-[(TCPP)Zn]-MOF	-55	200ns	49	65522ns		
Hf-[(TCPP)Zn]-MOF+phenol	-41	350ns	38	74280ns		
[(H_4TCPP)Zn] Ligand	-269	118ns	196	53532ns		
[(H_4TCPP)Zn] Ligand+0.01mM phenol	-273	127ns	202	52800ns		
[(H_4TCPP)Zn] Ligand+0.015mM phenol	-385	126ns	285	49612ns		
[(H_4TCPP)Zn] Ligand+0.04mM phenol	-314	103ns	230	53296ns		
[(H_4TCPP)Zn] Ligand+0.06mM phenol	-352	139ns	272	49364ns		
[(H_4TCPP)Zn] Ligand+0.1mM phenol	-123	124ns	92	65152ns		



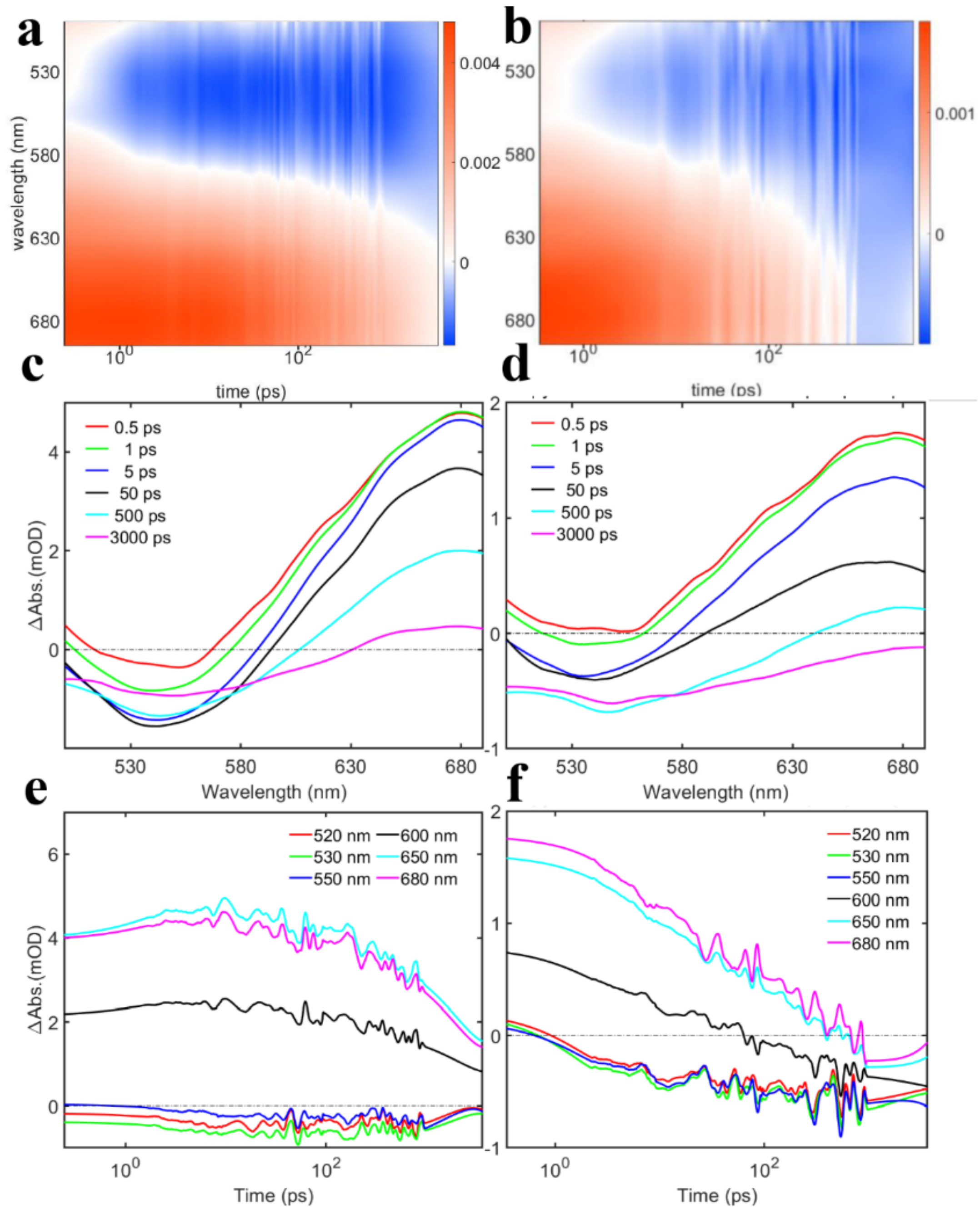
Supplementary Figure 55. Time traces of the nanosecond transient absorption spectra (400-500 nm) of solutions of [(H_4TCPP)Zn] ligand with different concentrations of phenol: 0 mM (a), 0.01mM (b), 0.015mM (c), 0.04mM (d), 0.06mM (e), 0.1mM (f). The excitation wavelength is 532nm.



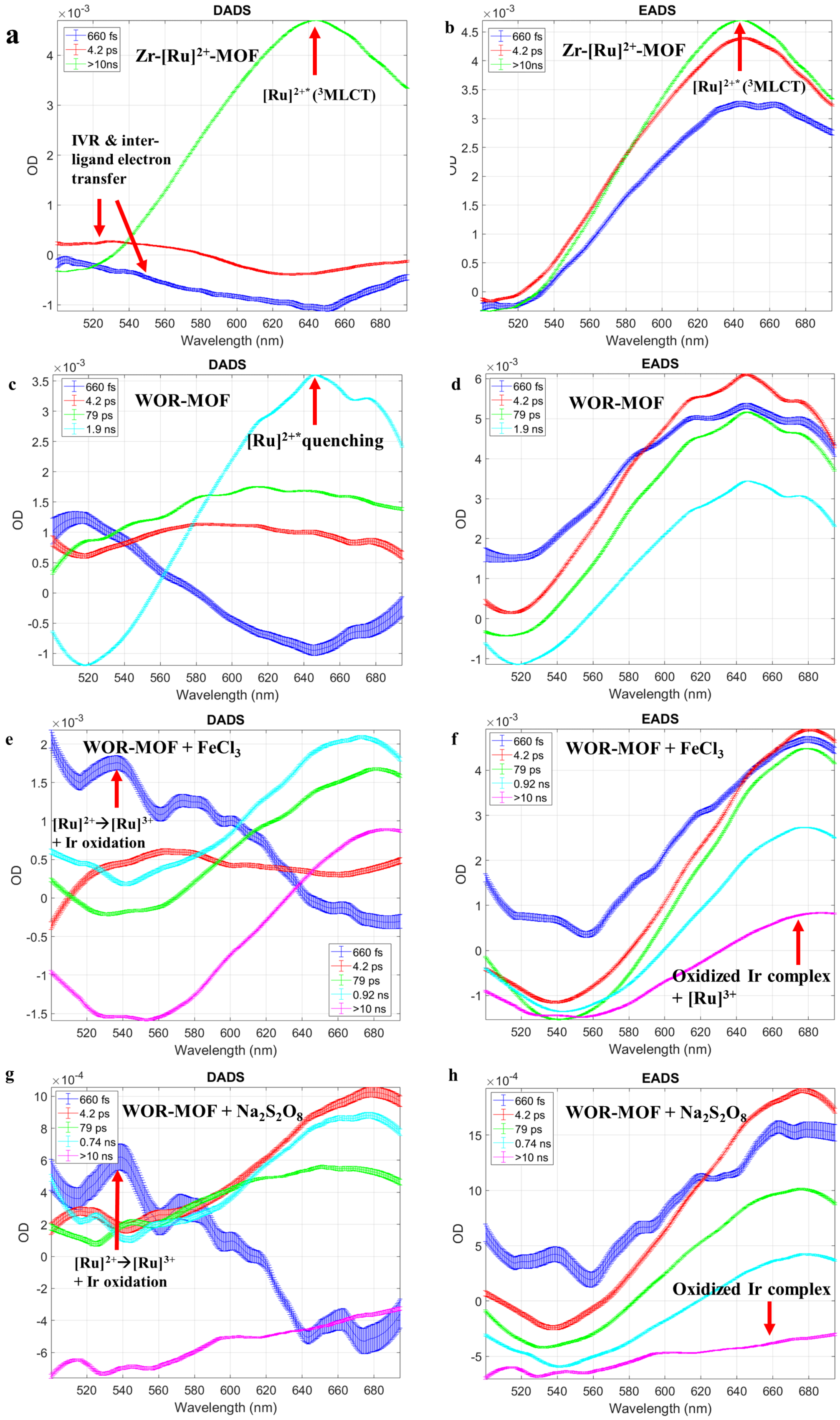
Supplementary Figure 56. The time trances of nanosecond transient absorption spectra of Hf-[(TCPP)Zn]-MOF (a), Hf-[(TCPP)Pt]-MOF (b), Hf-[(TCPP)Zn]-MOF+phenol (c), HER-MOF (d), HER-MOF+0.01 mM phenol (e), HER-MOF+0.015 mM phenol (f). The excitation wavelength is 532 nm.



Supplementary Figure 57. Transient absorption spectra of WOR-MOF (a,c,e) and Zr-$[Ru]^{2+}$-MOF (b,d,f).



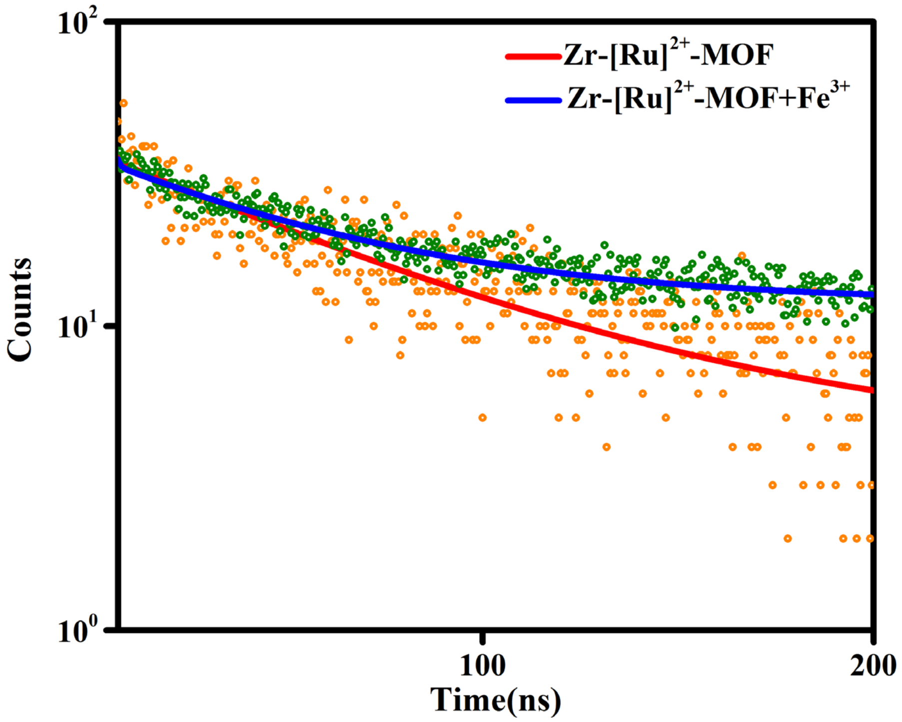
Supplementary Figure 58. Transient absorption spectra of WOR-MOF + $FeCl_3$ (a, c, e), and WOR-MOF + $Na_2S_2O_8$ (b, d, f).



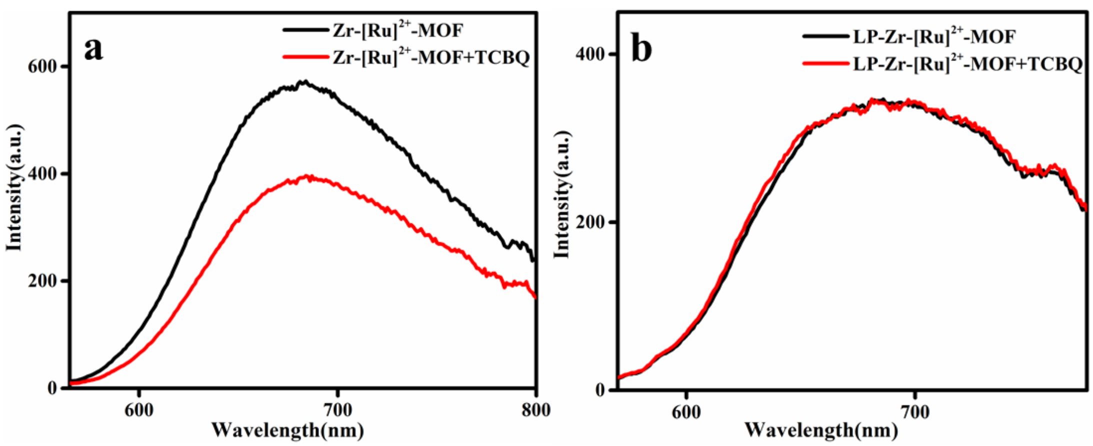
Supplementary Figure 59. DADS and EADS from global fittings of fsTA spectra of Zr-$[Ru]^{2+}$-MOF (a, b), WOR-MOF containing $[Ru]^{2+}$/[(BPYDC)Ir] (c, d), and WOR-MOF + $FeCl_3$ (e, f), WOR-MOF + $Na_2S_2O_8$ (g, h).

Supplementary Table 11. Time constants from global fitting of fsTA spectra of Zr-[$Ru^{II}(BPY)_2(BPYDC)$]-MOFs and homogeneous ligands

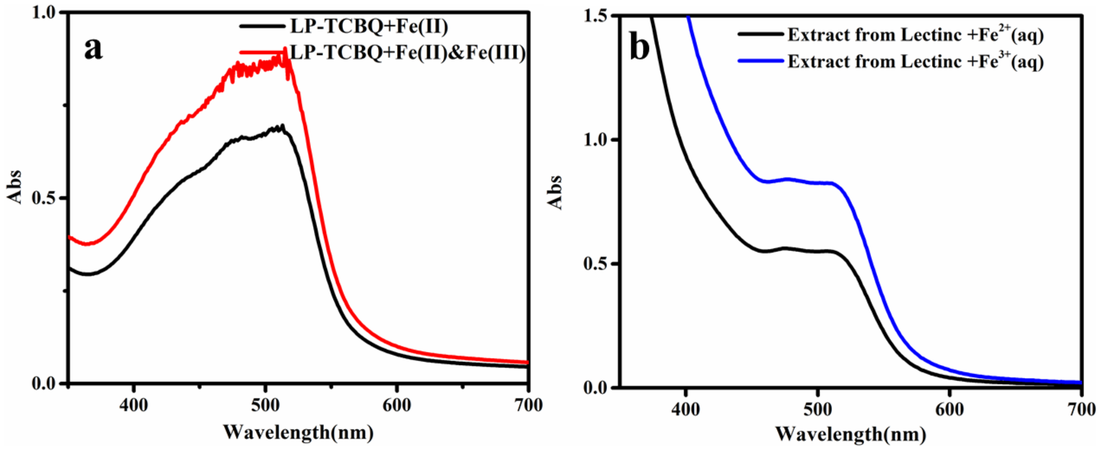
Sample	Time constants					
	1st	2nd	3rd	4th	5th	6th
Zr-[Ru]$^{2+}$-MOF	660fs	4.2ps				>10ns
WOR-MOF	660fs	4.2ps	79ps	1.9ns		
WOR-MOF + $FeCl_3$	660fs	4.2ps	79ps	0.92ns		>10ns
WOR-MOF + $Na_2S_2O_8$	660fs	4.2ps	79ps	0.74ns		>10ns



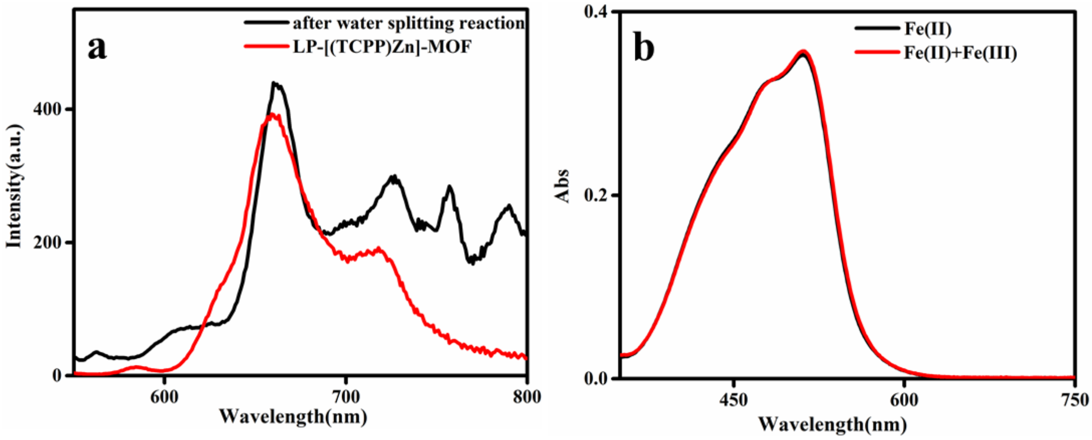
Supplementary Figure 60. Phosphorescence lifetime spectra of Zr-[Ru]$^{2+}$-MOF and Zr-[Ru]$^{2+}$-MOF+$FeCl_3$.



Supplementary Figure 61. Fluorescence quenching by adding TCBQ into the aqueous suspension of the Zr-[Ru]$^{2+}$-MOF(a) and the liposome hybrid LP-Zr-[Ru]$^{2+}$-MOF (b). The concentration of quencher is 1mM. We find that TCBQ cannot quench the emission of the LP-[Ru]$^{2+}$-MOF, which shows effective seperation between TCBQ (in the lipid bilayer) and the Zr-[Ru]$^{2+}$-MOF (in the aqueous phase) by the liposome structure.



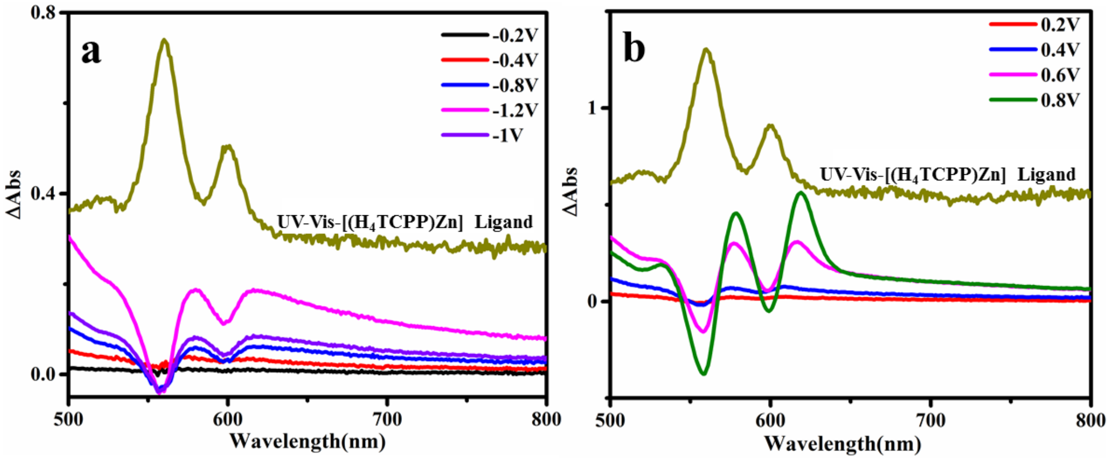
Supplementary Figure 62. UV-Vis spectra to determine the amount of Fe^{2+} and Fe^{3+} in the aqueous solution using 1,10-phenanthroline monohydrate as indicator after the reaction between Fe^{2+} in the aqueous solution and TCBQ in the lipid bilayer (a). The amount of Fe^{3+} was obtained by adding hydroxylamine hydrochloride as reducing agent to reduce the Fe^{3+} to Fe^{2+}. The UV-Vis spectra before the addition of hydroxylamine hydrochloride were due to Fe^{2+} in the analysis solution while the UV-Vis spectra after the addition of hydroxylamine hydrochloride correpsond to the sum of Fe^{2+} + Fe^{3+} of the original analysis solution. The reaction solution after the reaction between Fe^{2+} in the aquoues phase and TCBQ in the liposome bilayer is shown here, demonstrating that the TCBQ can effectively oxidize Fe^{2+} to Fe^{3+} even if they are not in the same phase. Adsorption experiment of Fe^{2+} and Fe^{3+} on Lecithin (b). After extracting aqueous solution of the Fe^{2+} and Fe^{3+} with the ether solution of Lecithin, the organic phase was added 1,10-phenanthroline monohydrate and/or hydroxylamine hydrochloride to detect Fe. The result shows that the polar phosphate group on the Lecithin can adsorb the Fe^{3+} and Fe^{2+}, which may facilitate the reaction between Fe^{2+} and TCBQ on the liposome/water interface.



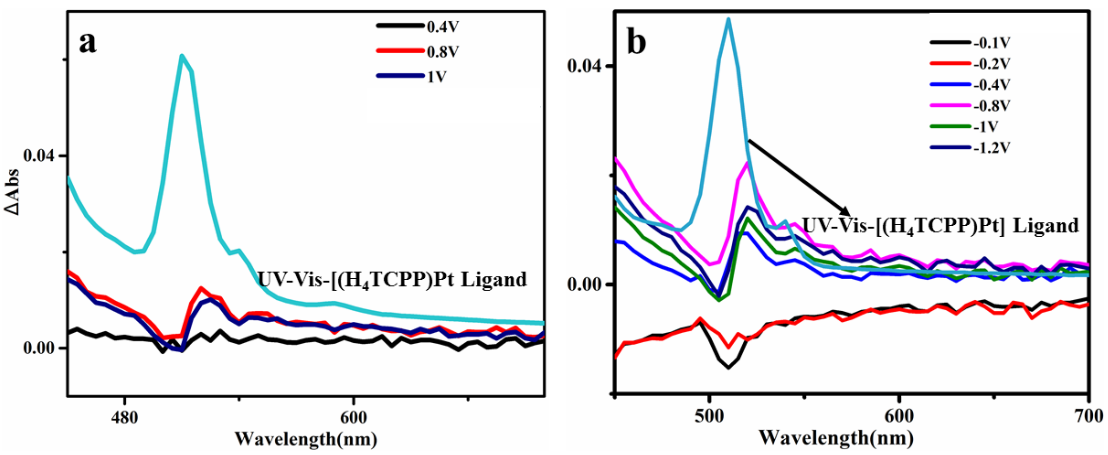
Supplementary Figure 63. Fluorescence spectra of LP-WOR-HER-MOF after reaction (a). The retention of the emission of [(TCPP)Zn] indicates that the resting state of the TCBQ in the reaction is in its reductive form HTCBQ, as the TCBQ can efficiently quench the [(TCPP)Zn] emission. The UV-Vis spectra to determine the amount of Fe^{2+} and Fe^{3+} in the aqueous phase after photocatalysis using LP-WOR-HER-MOF by adding 1,10-phenanthroline monohydrate and/or hydroxylamine hydrochloride (b). The results show that the majority of the Fe is in the Fe^{2+} oxidation state.

UV-Vis spectra under electrochemistry scan to produce difference spectra. For spectroelectrochemistry, solutions of analytes were prepared in DMF with 0.2 M of tetra-^{n}Bu-ammonium perchlorate electrolyte. Spectra were recorded with a Varian/Agilent Cary 5000 between 400 and 800 nm. A home-built three neck cell was used for the three-electrode setup with platinum gauze as working electrode, Pt wire as counter electrode, and Ag/AgCl as pseudo-reference electrode. Argon was used to flush the cell before (20 min) and during measurements to remove oxygen. Potentials were provided by a CHI1000C, controlled via CHI1000C Electro Chemical Analyzer software.

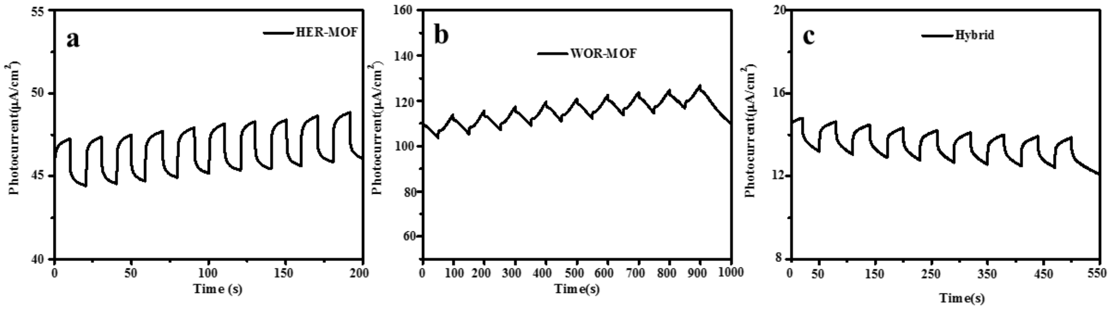
Electrochemical cyclic voltammetry experiment. For cyclic voltammetry, solutions of analytes were prepared in DMF with 0.2 M of tetra(n-butyl)ammonium perchlorate electrolyte. A home-built three neck cell was used for the three-electrode setup with a platinum electrode as working electrode, a Pt wire as counter electrode, and Ag/AgCl as pseudo-reference electrode.



Supplementary Figure 64. Differential absorption spectra of [(H_4TCPP)Zn] ligand under reduction potential (a) and oxidation potential (b).



Supplementary Figure 65. Differential absorption spectra of [(H_4TCPP)Pt] Ligand under oxidation potential (a) and reduction potential (b).

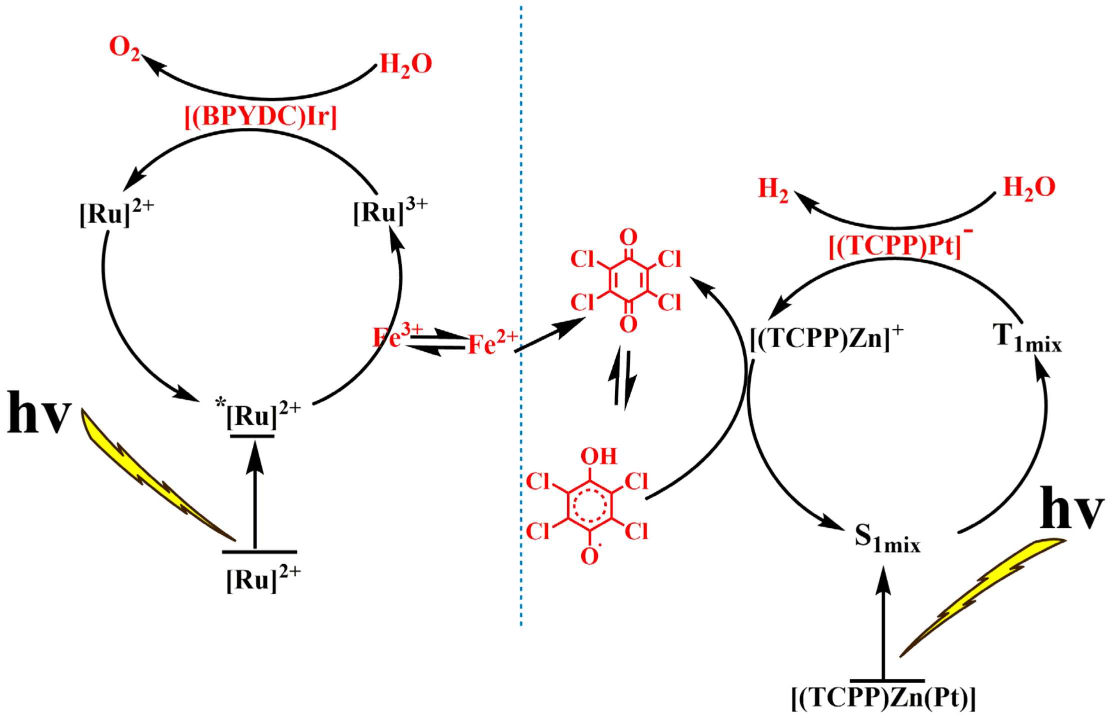


Supplementary Figure 66. Photocurrent responses for HER-MOF, WOR-MOF and hybrid.

Supplementary Table 12. Solution redox potentials of the ligands and redox shutters in the total water splitting system vs NHE[a]

Sample	Oxidation(V)		Reduction(V)		Ref.
	1st	2nd	1st	2nd	
[(H_4TCPP)Zn] Ligand	1.22		-0.75	-1.72	This work[18]
[(H_4TCPP)Pt] Ligand	1.27		-0.75	-1.60	This work
[(BPYDC)Ir]	1.34				[19]
$[Ru]^{2+}$ Ligand	1.47		-1.3	-1.47	This work
$FeCl_3$	0.77				[20]
TCBQ	0.80		-0.065		This work[21]

[a]All Cyclic voltammograms were measured at a scan rate of 100 mV s^{-1} using a Pt electrode as working electrode in DMF Solution.



Supplementary Figure 67. Proposed catalytic mechanism for the total water splitting.

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